

OIL RECOVERY MECHANISMS AND THE MATERIAL BALANCE EQUATION

Each reservoir is composed of a unique combination of geometric form, geological rock properties, fluid characteristics, and primary drive mechanism. Although no two reservoirs are identical in all aspects, they can be grouped according to the primary recovery mechanism by which they produce. It has been observed that each drive mechanism has certain typical performance characteristics in terms of:

- Ultimate recovery factor
- Pressure decline rate
- Gas-oil ratio
- Water production

The recovery of oil by any of the natural drive mechanisms is called **primary recovery**. The term refers to the production of hydrocarbons from a reservoir without the use of any process (such as fluid injection) to supplement the natural energy of the reservoir.

The two main objectives of this chapter are to:

1. Introduce and give a detailed discussion of the various primary recovery mechanisms and their effects on the overall performance of oil reservoirs.

2. Provide the basic principles of the material balance equation and other governing relationships that can be used to predict the volumetric performance of oil reservoirs.

PRIMARY RECOVERY MECHANISMS

For a proper understanding of reservoir behavior and predicting future performance, it is necessary to have knowledge of the driving mechanisms that control the behavior of fluids within reservoirs. The overall performance of oil reservoirs is largely determined by the nature of the energy, i.e., driving mechanism, available for moving the oil to the well-bore. There are basically six driving mechanisms that provide the natural energy necessary for oil recovery:

- Rock and liquid expansion drive
- Depletion drive
- Gas-cap drive
- Water drive
- Gravity drainage drive
- Combination drive

These driving mechanisms are discussed as follows.

Rock and Liquid Expansion

When an oil reservoir initially exists at a pressure higher than its bubble-point pressure, the reservoir is called an **undersaturated-oil reservoir**. At pressures above the bubble-point pressure, crude oil, connate-water, and rock are the only materials present. As the reservoir pressure declines, the rock and fluids expand due to their individual compressibilities. The reservoir rock compressibility is the result of two factors:

- Expansion of the individual rock grains
- Formation compaction

Both of the above two factors are the results of a decrease of fluid pressure within the pore spaces, and both tend to reduce the pore volume through the reduction of the porosity.

As the expansion of the fluids and reduction in the pore volume occur with decreasing reservoir pressure, the crude oil and water will be forced

out of the pore space to the wellbore. Because liquids and rocks are only slightly compressible, the reservoir will experience a rapid pressure decline. The oil reservoir under this driving mechanism is characterized by a constant gas-oil ratio that is equal to the gas solubility at the bubble point pressure.

This driving mechanism is considered the least efficient driving force and usually results in the recovery of only a small percentage of the total oil-in-place.

The Depletion-Drive Mechanism

This driving form may also be referred to by the following various terms:

- Solution gas drive
- Dissolved gas drive
- Internal gas drive

In this type of reservoir, the principal source of energy is a result of gas liberation from the crude oil and the subsequent expansion of the solution gas as the reservoir pressure is reduced. As pressure falls below the bubble-point pressure, gas bubbles are liberated within the microscopic pore spaces. These bubbles expand and force the crude oil out of the pore space as shown conceptually in Figure 11-1.

Cole (1969) suggests that a depletion-drive reservoir can be identified by the following characteristics:

- **Reservoir pressure:** The reservoir pressure declines rapidly and continuously. This reservoir pressure behavior is attributed to the fact that no extraneous fluids or gas caps are available to provide a replacement of the gas and oil withdrawals.
- **Water production:** The absence of a water drive means there will be little or no water production with the oil during the entire producing life of the reservoir.
- **Gas-oil ratio:** A depletion-drive reservoir is characterized by a rapidly increasing gas-oil ratio from all wells, regardless of their structural position. After the reservoir pressure has been reduced below the bubble-point pressure, gas evolves from solution throughout the reservoir. Once the gas saturation exceeds the critical gas saturation, free gas begins to flow toward the wellbore and the gas-oil ratio increases. The

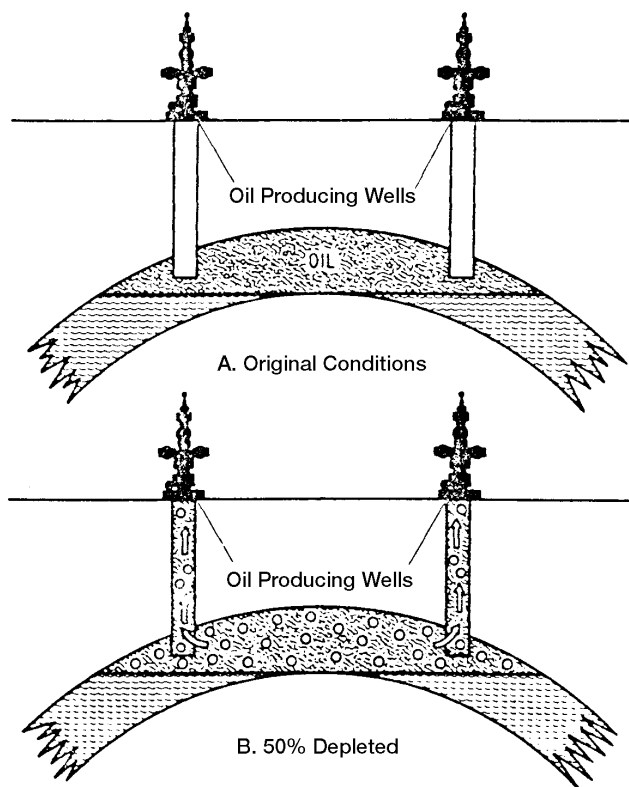


Figure 11-1. Solution-gas-drive reservoir. (After Clark, N. J., *Elements of Petroleum Reservoirs*, SPE, 1969.)

gas will also begin a vertical movement due to the gravitational forces, which may result in the formation of a secondary gas cap. Vertical permeability is an important factor in the formation of a secondary gas cap.

- **Ultimate oil-recovery:** Oil production by depletion drive is usually the least efficient recovery method. This is a direct result of the formation of gas saturation throughout the reservoir. Ultimate oil recovery from depletion-drive reservoirs may vary from less than 5% to about 30%. The low recovery from this type of reservoirs suggests that large quantities of oil remain in the reservoir and, therefore, depletion-drive reservoirs are considered the best candidates for secondary recovery applications.

The above characteristic trends occurring during the production life of depletion-drive reservoirs are shown in Figure 11-2 and summarized below:

Characteristics	Trend
Reservoir pressure	Declines rapidly and continuously
Gas-oil ratio	Increases to maximum and then declines
Water production	None
Well behavior	Requires pumping at early stage
Oil recovery	5% to 30%

Gas-Cap Drive

Gas-cap-drive reservoirs can be identified by the presence of a gas cap with little or no water drive as shown in Figure 11-3.

Due to the ability of the gas cap to expand, these reservoirs are characterized by a slow decline in the reservoir pressure. The natural energy available to produce the crude oil comes from the following two sources:

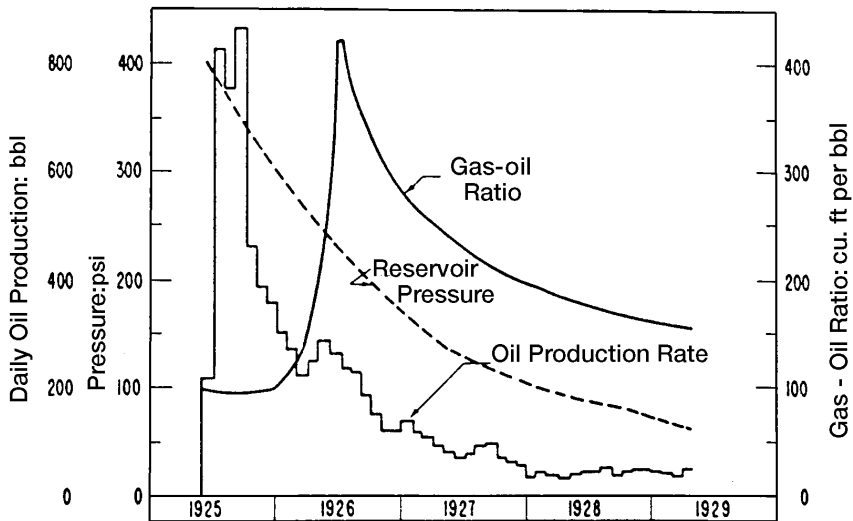


Figure 11-2. Production data of a solution-gas-drive reservoir. (After Clark, N. J., *Elements of Petroleum Reservoirs*, SPE, 1969.)

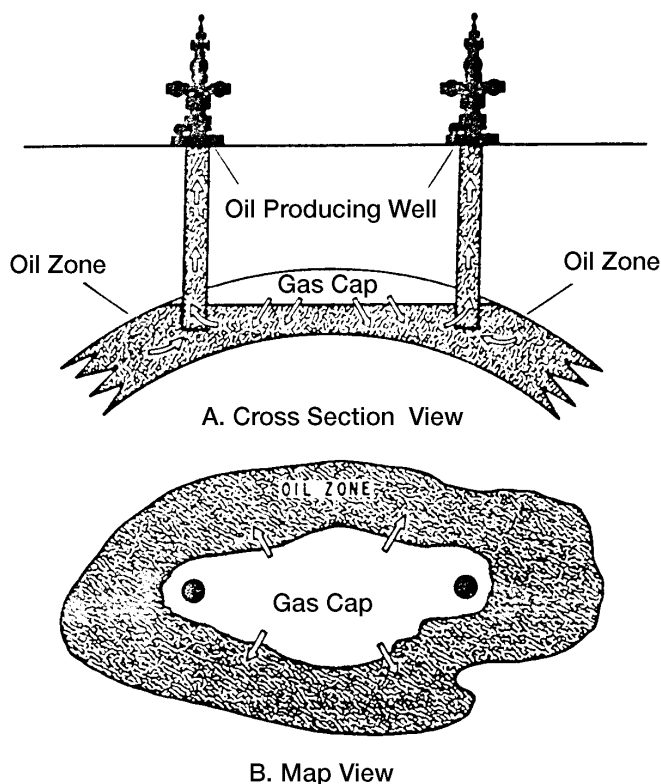


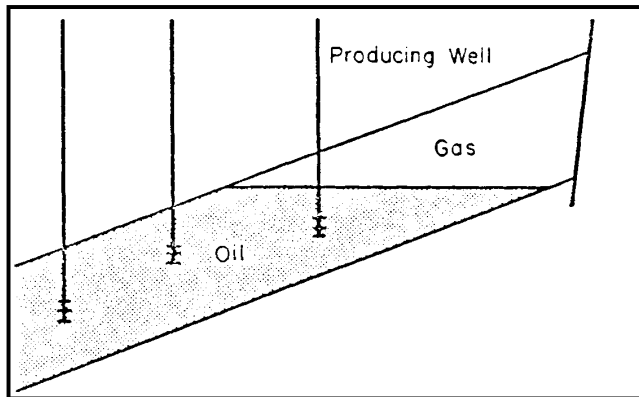
Figure 11-3. Gas-cap-drive reservoir. (After Clark, N. J., *Elements of Petroleum Reservoirs, SPE, 1969.*)

- Expansion of the gas-cap gas
- Expansion of the solution gas as it is liberated

Cole (1969) and Clark (1969) presented a comprehensive review of the characteristic trends associated with gas-cap-drive reservoirs. These characteristic trends are summarized below:

- **Reservoir pressure:** The reservoir pressure falls slowly and continuously. Pressure tends to be maintained at a higher level than in a depletion-drive reservoir. The degree of pressure maintenance depends upon the volume of gas in the gas cap compared to the oil volume.
- **Water production:** Absent or negligible water production.

- **Gas-oil ratio:** The gas-oil ratio rises continuously in up-structure wells. As the expanding gas cap reaches the producing intervals of upstructure wells, the gas-oil ratio from the affected wells will increase to high values.
- **Ultimate oil recovery:** Oil recovery by gas-cap expansion is actually a frontal drive displacing mechanism that, therefore, yields a considerably larger recovery efficiency than that of depletion-drive reservoirs. This larger recovery efficiency is also attributed to the fact that no gas saturation is being formed throughout the reservoir at the same time. Figure 11-4 shows the relative positions of the gas-oil contact at different times in the producing life of the reservoir. The expected oil recovery ranges from 20% to 40%.



A. Initial fluid distribution.

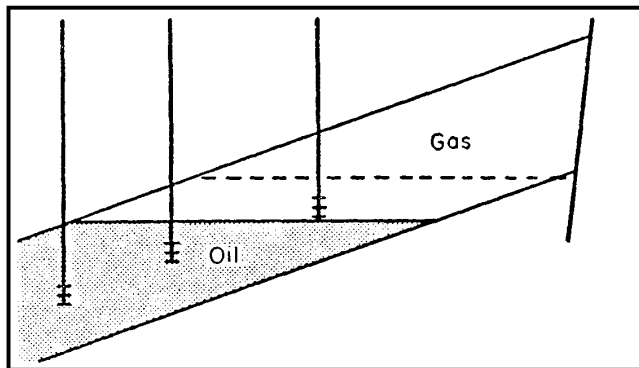


Figure 11-4. Gas-cap-drive reservoir. (After Cole, F., *Reservoir Engineering Manual*, Gulf Publishing Company, 1969.)

- **Well behavior:** Because of effects of gas-cap expansion on maintaining reservoir pressure and the effect of decreased liquid column weight as it is produced out of the well, gas-cap-drive reservoirs tend to flow longer than depletion-drive reservoirs.

The ultimate oil recovery from a gas-cap-drive reservoir will vary depending largely on the following six important parameters:

Size of the Original Gas Cap

As shown graphically in Figure 11-5, the ultimate oil recovery increases with increasing the size of the gas cap.

Vertical Permeability

Good vertical permeability will permit the oil to move downward with less bypassing of gas.

Oil Viscosity

As the oil viscosity increases, the amount of gas bypassing will also increase, which leads to a lower oil recovery.

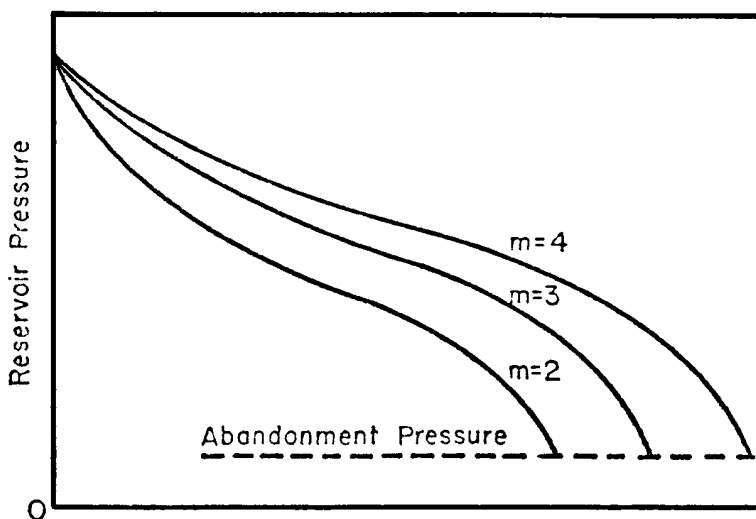


Figure 11-5. Effect of gas-cap size on ultimate oil recovery. (After Cole, F., Reservoir Engineering Manual, Gulf Publishing Company, 1969.)

Degree of Conservation of the Gas

In order to conserve gas, and thereby increase ultimate oil recovery, it is necessary to shut-in the wells that produce excessive gas.

Oil Production Rate

As the reservoir pressure declines with production, solution gas evolves from the crude oil and the gas saturation increases continuously. If the gas saturation exceeds the critical gas saturation, the evolved gas begins to flow in the oil zone. As a result of creating a mobile gas phase in the oil zone, the following two events will occur:

- The effective permeability to oil will be decreased as a result of the increased gas saturation.
- The effective permeability to gas will be increased, thereby increasing the flow of gas.

The formation of the free gas saturation in the oil zone cannot be prevented without resorting to pressure maintenance operations. Therefore, in order to achieve maximum benefit from a gas-cap-drive-producing mechanism, gas saturation in the oil zone must be kept to an absolute minimum. This can be accomplished by taking advantage of gravitational segregation of the fluids. In fact, an efficiently operated gas-cap-drive reservoir must also have an efficient gravity segregation drive. As the gas saturation is formed in the oil zone it must be allowed to migrate upstructure to the gas cap. Thus, a gas-cap-drive reservoir is in reality a combination-driving reservoir, although it is not usually considered as such.

Lower producing rates will permit the maximum amount of free gas in the oil zone to migrate to the gas cap. Therefore, gas-cap-drive reservoirs are rate sensitive, as lower producing rates will usually result in increased recovery.

Dip Angle

The size of the gas cap, a measure of reservoir energy available to produce the oil, will in large part determine the recovery percent to be expected. Such recovery normally will be 20% to 40% of the original oil-in-place; if some other features are present to assist, however, such as a steep angle of dip that allows good oil drainage to the bottom of the

structure, considerably higher recoveries (up to 60% or greater) may be obtained. Conversely, extremely thin oil columns (where early breakthrough of the advancing gas cap occurs in producing wells) may limit oil recovery to lower figures regardless of the size of the gas cap. Figure 11-6 shows a typical production and pressure data for a gas-cap-drive reservoir.

The Water-Drive Mechanism

Many reservoirs are bounded on a portion or all of their peripheries by water bearing rocks called aquifers. The aquifers may be so large compared to the reservoir they adjoin as to appear infinite for all practical purposes, and they may range down to those so small as to be negligible in their effects on the reservoir performance.

The aquifer itself may be entirely bounded by impermeable rock so that the reservoir and aquifer together form a closed (volumetric) unit. On the other hand, the reservoir may be outcropped at one or more places where it may be replenished by surface water as shown schematically in Figure 11-7.

It is common to speak of edge water or bottom water in discussing water influx into a reservoir. Bottom water occurs directly beneath the oil and edge water occurs off the flanks of the structure at the edge of the oil

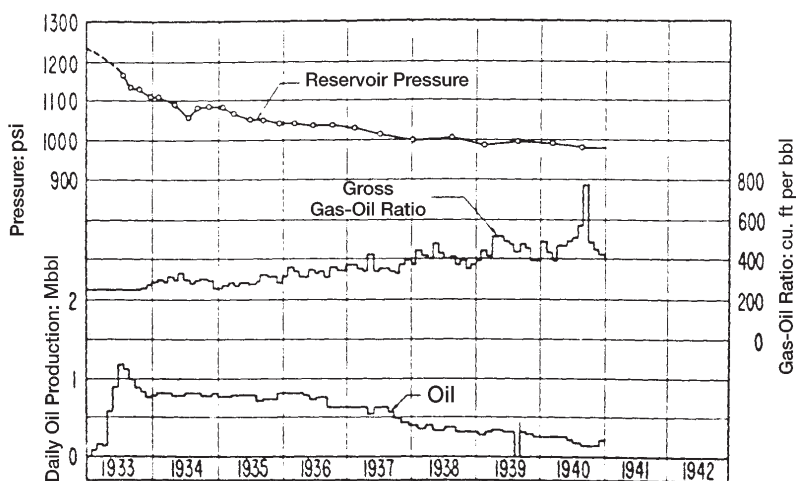


Figure 11-6. Production data for a gas-cap-drive reservoir. (After Clark, N. J., Elements of Petroleum Reservoirs, SPE, 1969. Courtesy of API.)

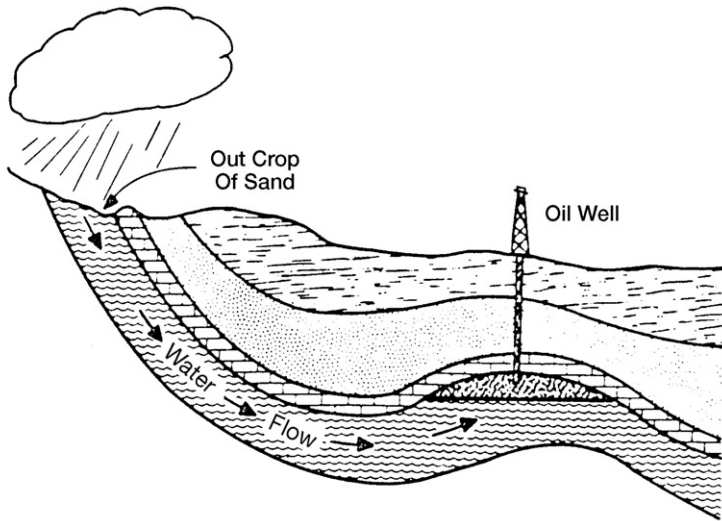


Figure 11-7. Reservoir having artesian water drive. (After Clark, N. J., *Elements of Petroleum Reservoirs*, SPE, 1969.)

as illustrated in Figure 11-8. Regardless of the source of water, the water drive is the result of water moving into the pore spaces originally occupied by oil, replacing the oil and displacing it to the producing wells.

Cole (1969) presented the following discussion on the characteristics that can be used for identification of the water-driving mechanism:

Reservoir Pressure

The reservoir pressure decline is usually very gradual. Figure 11-9 shows the pressure-production history of a typical water-drive reservoir.

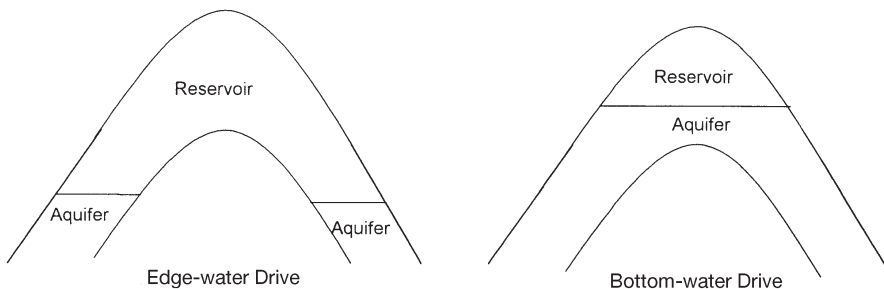


Figure 11-8. Aquifer geometries.

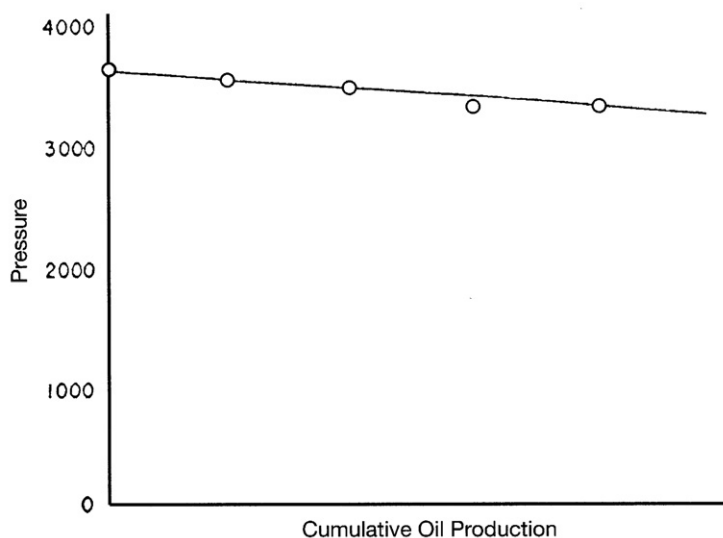


Figure 11-9. Pressure-production history for a water-drive reservoir.

It is not uncommon for many thousands of barrels of oil to be produced for each pound per square inch drop in reservoir pressure. The reason for the small decline in reservoir pressure is that oil and gas withdrawals from the reservoir are replaced almost volume for volume by water encroaching into the oil zone.

Several large oil reservoirs in the Gulf Coast areas of the United States have such active water drives that the reservoir pressure has declined only about 1 psi per million barrels of oil produced. Although pressure history is normally plotted versus cumulative oil production, it should be understood that total reservoir fluid withdrawals are the really important criteria in the maintenance of reservoir pressure. In a water-drive reservoir, only a certain number of barrels of water can move into the reservoir as a result of a unit pressure drop within the reservoir.

Since the principal income production is from oil, if the withdrawals of water and gas can be minimized, then the withdrawal of oil from the reservoir can be maximized with minimum pressure decline. Therefore, it is extremely important to reduce water and gas production to an absolute minimum. This can usually be accomplished by shutting in wells producing large quantities of these fluids and, where possible, transferring their allowables to other wells producing with lower water-oil or gas-oil ratios.

Water Production

Early excess water production occurs in structurally low wells. This is characteristic of a water-drive reservoir, and, provided the water is encroaching in a uniform manner, nothing can or should be done to restrict this encroachment, as the water will probably provide the most efficient displacing mechanism possible.

If the reservoir has one or more lenses of very high permeability, then the water may be moving through this more permeable zone. In this case, it may be economically feasible to perform remedial operations to shut off this permeable zone producing water. It should be realized that in most cases the oil that is being recovered from a structurally low well will be recovered from wells located higher on the structure and any expenses involved in remedial work to reduce the water-oil ratio of structurally low wells may be needless expenditures.

Gas-Oil Ratio

There is normally little change in the producing gas-oil ratio during the life of the reservoir. This is especially true if the reservoir does not have an initial free gas cap. Pressure will be maintained as a result of water encroachment and therefore there will be relatively little gas released from this solution.

Ultimate Oil Recovery

Ultimate recovery from water-drive reservoirs is usually much larger than recovery under any other producing mechanism. Recovery is dependent upon the efficiency of the flushing action of the water as it displaces the oil. In general, as the reservoir heterogeneity increases, the recovery will decrease, due to the uneven advance of the displacing water.

The rate of water advance is normally faster in the zones of high permeability. This results in earlier high water-oil ratios and consequent earlier economic limits. Where the reservoir is more or less homogeneous, the advancing waterfront will be more uniform, and when the economic limit, due primarily to high water-oil ratio, has been reached, a greater portion of the reservoir will have been contacted by the advancing water.

Ultimate oil recovery is also affected by the degree of activity of the water drive. In a very active water drive where the degree of pressure maintenance is good, the role of solution gas in the recovery process is

reduced to almost zero, with maximum advantage being taken of the water as a displacing force. This should result in maximum oil recovery from the reservoir. The ultimate oil recovery normally ranges from 35% to 75% of the original oil-in-place.

The characteristic trends of a water-drive reservoir are shown graphically in Figure 11-10 and are summarized below:

Characteristics	Trends
Reservoir pressure	Remains high
Surface gas-oil ratio	Remains low
Water production	Starts early and increases to appreciable amounts
Well behavior	Flow until water production gets excessive
Expected oil recovery	35% to 75%

The Gravity-Drainage-Drive Mechanism

The mechanism of gravity drainage occurs in petroleum reservoirs as a result of differences in densities of the reservoir fluids. The effects of gravitational forces can be simply illustrated by placing a quantity of crude oil and a quantity of water in a jar and agitating the contents. After agitation, the jar is placed at rest, and the denser fluid (normally water)

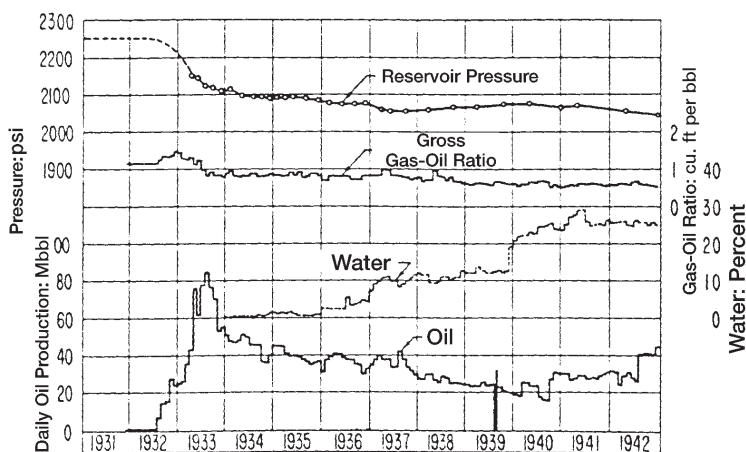


Figure 11-10. Production data for a water-drive reservoir. (After Clark, N. J., Elements of Petroleum Reservoirs, SPE, 1969. Courtesy of API.)

will settle to the bottom of the jar, while the less dense fluid (normally oil) will rest on top of the denser fluid. The fluids have separated as a result of the gravitational forces acting on them.

The fluids in petroleum reservoirs have all been subjected to the forces of gravity, as evidenced by the relative positions of the fluids, i.e., gas on top, oil underlying the gas, and water underlying oil. The relative positions of the reservoir fluids are shown in Figure 11-11. Due to the long periods of time involved in the petroleum accumulation-and-migration process, it is generally assumed that the reservoir fluids are in equilibrium. If the reservoir fluids are in equilibrium, then the gas-oil and oil-water contacts should be essentially horizontal. Although it is difficult to determine precisely the reservoir fluid contacts, best available data indicate that, in most reservoirs, the fluid contacts actually are essentially horizontal.

Gravity segregation of fluids is probably present to some degree in all petroleum reservoirs, but it may contribute substantially to oil production in some reservoirs.

Cole (1969) stated that reservoirs operating largely under a gravity-drainage-producing mechanism are characterized by:

- Reservoir Pressure

Variable rates of pressure decline, depending principally upon the amount of gas conservation. Strictly speaking, where the gas is conserved and reservoir pressure is maintained, the reservoir would be operating

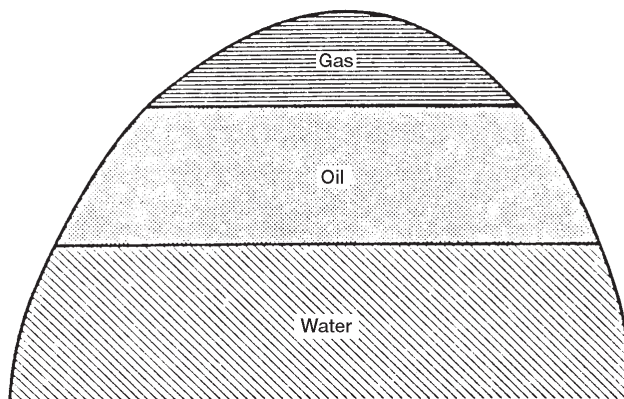


Figure 11-11. Initial fluids distribution in an oil reservoir.

under combined gas-cap drive and gravity-drainage mechanisms. Therefore, for the reservoir to be operating solely as a result of gravity drainage, the reservoir would show a rapid pressure decline. This would require the upstructure migration of the evolved gas where it later would be produced from structurally high wells, resulting in rapid loss of pressure.

- **Gas-Oil Ratio**

Low gas-oil ratio from structurally low wells. This is caused by migration of the evolved gas upstructure due to gravitational segregation of the fluids. On the other hand, the structurally high wells will experience an increasing gas-oil ratio as a result of the upstructure migration of the gas released from the crude oil.

- **Secondary Gas Cap**

Formation of a secondary gas cap in reservoirs that initially were undersaturated. Obviously the gravity-drainage mechanism does not become operative until reservoir pressure has declined below the saturation pressure, since above the saturation pressure there will be no free gas in the reservoir.

- **Water Production**

Little or no water production. Water production is indicative of a water drive.

Ultimate Oil Recovery

Ultimate recovery from gravity-drainage reservoirs will vary widely, due primarily to the extent of depletion by gravity drainage alone. Where gravity drainage is good, or where producing rates are restricted to take maximum advantage of the gravitational forces, recovery will be high. There are reported cases where recovery from gravity-drainage reservoirs has exceeded 80% of the initial oil-in-place. In other reservoirs where depletion drive also plays an important role in the oil recovery process, the ultimate recovery will be less.

In operating a gravity-drainage reservoir, it is essential that the oil saturation in the vicinity of the wellbore must be maintained as high as possible. There are two basic reasons for this requirement:

- A high oil saturation means a higher oil flow rate
- A high oil saturation means a lower gas flow rate

If the evolved gas migrates upstructure instead of toward the wellbore, then a high oil saturation in the vicinity of the wellbore can be maintained.

In order to take maximum advantage of the gravity-drainage-producing mechanism, wells should be located as structurally low as possible. This will result in maximum conservation of the reservoir gas. A typical gravity-drainage reservoir is shown in Figure 11-12.

Factors that affect ultimate recovery from gravity-drainage reservoirs are:

- Permeability in the direction of dip
- Dip of the reservoir
- Reservoir producing rates
- Oil viscosity
- Relative permeability characteristics

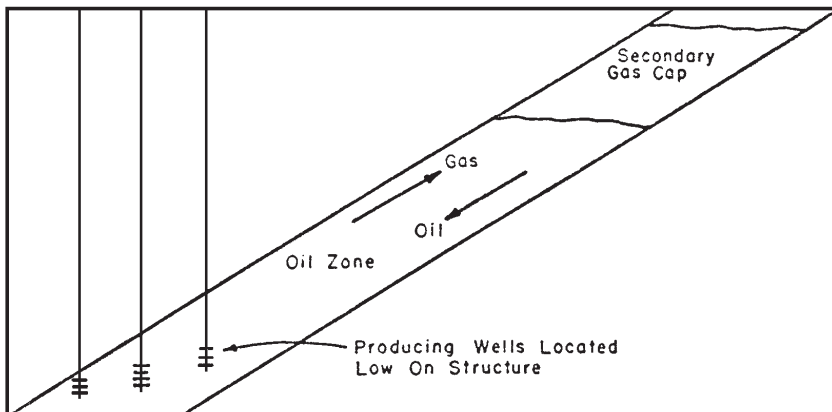


Figure 11-12. Gravity-drainage reservoir. (After Cole, F., *Reservoir Engineering Manual*, Gulf Publishing Company, 1969.)

Cole (1969) presented the following complete treatment of the above listed factors.

Permeability in the Direction of Dip

Good permeability in the direction of migration of the oil is a prerequisite for efficient gravity drainage. For example, a reservoir with little structural relief that also contained many more or less continuous shale “breaks” could probably not be operated under gravity drainage because the oil could not flow to the base of the structure.

Dip of the Reservoir

In most reservoirs, the permeability in the direction of dip is considerably larger than the permeability transverse to the direction of dip. Therefore, as the dip of the reservoir increases, the oil and gas can flow along the direction of dip (which is also the direction of greatest permeability) and still achieve their desired structural position.

Reservoir-Producing Rates

Since the gravity-drainage rate is limited, the reservoir-producing rates should be limited to the gravity-drainage rate, and then maximum recovery will result. If the reservoir-producing rate exceeds the gravity-drainage rate, the depletion-drive-producing mechanism will become more significant with a consequent reduction in ultimate oil recovery.

Oil Viscosity

Oil viscosity is important because the gravity-drainage rate is dependent upon the viscosity of the oil. In the fluid flow equations, the flow rate increases as the viscosity decreases. Therefore, the gravity-drainage rate will increase as the reservoir oil viscosity decreases.

Relative Permeability Characteristics

For an efficient gravity-drive mechanism to be operative, the gas must flow upstructure while the oil flows downstructure. Although this situation involves counterflow of the oil and gas, both fluids are flowing and, therefore, relative permeability characteristics of the formation are very important.

The Combination-Drive Mechanism

The driving mechanism most commonly encountered is one in which both water and free gas are available in some degree to displace the oil toward the producing wells. The most common type of drive encountered, therefore, is a combination-drive mechanism as illustrated in Figure 11-13.

Two combinations of driving forces can be present in combination-drive reservoirs. These are (1) depletion drive and a weak water drive and; (2) depletion drive with a small gas cap and a weak water drive. Then, of course, gravity segregation can play an important role in any of the aforementioned drives.

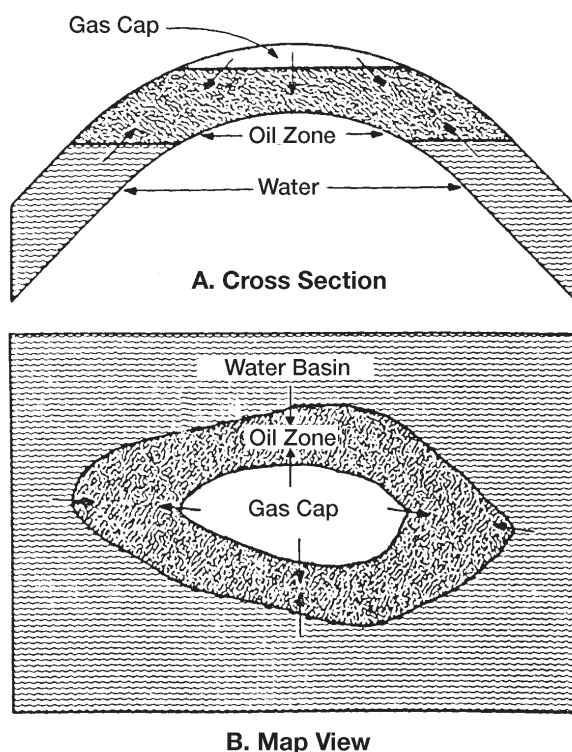


Figure 11-13. Combination-drive reservoir. (After Clark, N. J., *Elements of Petroleum Reservoirs*, SPE, 1969.)

Combination-drive reservoirs can be recognized by the occurrence of a combination of some of the following factors:

- a. Relatively rapid pressure decline. Water encroachment and/or external gas-cap expansion are insufficient to maintain reservoir pressures.
- b. Water encroaching slowly into the lower part of the reservoir. Structurally low producing wells will exhibit slowly increasing water producing rates.
- c. If a small gas cap is present the structurally high wells will exhibit continually increasing gas-oil ratios, provided the gas cap is expanding. It is possible that the gas cap will shrink due to production of excess free gas, in which case the structurally high wells will exhibit a **decreasing** gas-oil ratio. This condition should be avoided whenever possible, as large volumes of oil can be lost as a result of a shrinking gas cap.
- d. A substantial percentage of the total oil recovery may be due to the depletion-drive mechanism. The gas-oil ratio of structurally low wells will also continue to increase due to evolution of solution gas throughout the reservoir, as pressure is reduced.
- e. Ultimate recovery from combination-drive reservoirs is usually greater than recovery from depletion-drive reservoirs but less than recovery from water-drive or gas-cap-drive reservoirs. Actual recovery will depend upon the degree to which it is possible to reduce the magnitude of recovery by depletion drive. In most combination-drive reservoirs, it will be economically feasible to institute some type of pressure maintenance operation, either gas injection, water injection, or both gas and water injection, depending upon the availability of the fluids.

THE MATERIAL BALANCE EQUATION

The material balance equation (MBE) has long been recognized as one of the basic tools of reservoir engineers for interpreting and predicting reservoir performance. The MBE, when properly applied, can be used to:

- Estimate initial hydrocarbon volumes in place
- Predict future reservoir performance
- Predict ultimate hydrocarbon recovery under various types of primary driving mechanisms

The equation is structured to simply keep inventory of all materials entering, leaving, and accumulating in the reservoir. The concept of the material balance equation was presented by Schilthuis in 1941. In its simplest form, the equation can be written on a volumetric basis as:

$$\text{Initial volume} = \text{volume remaining} + \text{volume removed}$$

Since oil, gas, and water are present in petroleum reservoirs, the material balance equation can be expressed for the total fluids or for any one of the fluids present.

Before deriving the material balance, it is convenient to denote certain terms by symbols for brevity. The symbols used conform where possible to the standard nomenclature adopted by the Society of Petroleum Engineers.

p_i	Initial reservoir pressure, psi
p	Volumetric average reservoir pressure
Δp	Change in reservoir pressure = $p_i - p$, psi
p_b	Bubble point pressure, psi
N	Initial (original) oil-in-place, STB
N_p	Cumulative oil produced, STB
G_p	Cumulative gas produced, scf
W_p	Cumulative water produced, bbl
R_p	Cumulative gas-oil ratio, scf/STB
GOR	Instantaneous gas-oil ratio, scf/STB
R_{si}	Initial gas solubility, scf/STB
R_s	Gas solubility, scf/STB
B_{oi}	Initial oil formation volume factor, bbl/STB
B_o	Oil formation volume factor, bbl/STB
B_{gi}	Initial gas formation volume factor, bbl/scf
B_g	Gas formation volume factor, bbl/scf
W_{inj}	Cumulative water injected, STB
G_{inj}	Cumulative gas injected, scf
W_e	Cumulative water influx, bbl
m	Ratio of initial gas-cap gas reservoir volume to initial reservoir oil volume, bbl/bbl
G	Initial gas-cap gas, scf
P.V	Pore volume, bbl
c_w	Water compressibility, psi^{-1}
c_f	Formation (rock) compressibility, psi^{-1}

Several of the material balance calculations require the total pore volume (P.V) as expressed in terms of the initial oil volume N and the volume of the gas cap. The expression for the total pore volume can be

derived by conveniently introducing the parameter m into the relationship as follows:

Defining the ratio m as:

$$m = \frac{\text{Initial volume of gas cap}}{\text{Volume of oil initially in place}} = \frac{G B_{gi}}{N B_{oi}}$$

Solving for the volume of the gas cap gives:

$$\text{Initial volume of the gas cap} = G B_{gi} = m N B_{oi}$$

The total volume of the hydrocarbon system is then given by:

$$\text{Initial oil volume} + \text{initial gas cap volume} = (P.V) (1 - S_{wi})$$

$$N B_{oi} + m N B_{oi} = (P.V) (1 - S_{wi})$$

or

$$P.V = \frac{N B_{oi} (1 + m)}{1 - S_{wi}} \quad (11-1)$$

where S_{wi} = initial water saturation

N = initial oil-in-place, STB

$P.V$ = total pore volume, bbl

m = ratio of initial gas-cap gas reservoir volume to initial reservoir oil volume, bbl/bbl

Treating the reservoir pore as an idealized container as illustrated in Figure 11-14, volumetric balance expressions can be derived to account for all volumetric changes which occur during the natural productive life of the reservoir.

The MBE can be written in a generalized form as follows:

$$\begin{aligned} &\text{Pore volume occupied by the oil initially in place at } p_i \\ &\quad + \\ &\text{Pore volume occupied by the gas in the gas cap at } p_i \\ &\quad = \\ &\text{Pore volume occupied by the remaining oil at } p \end{aligned}$$

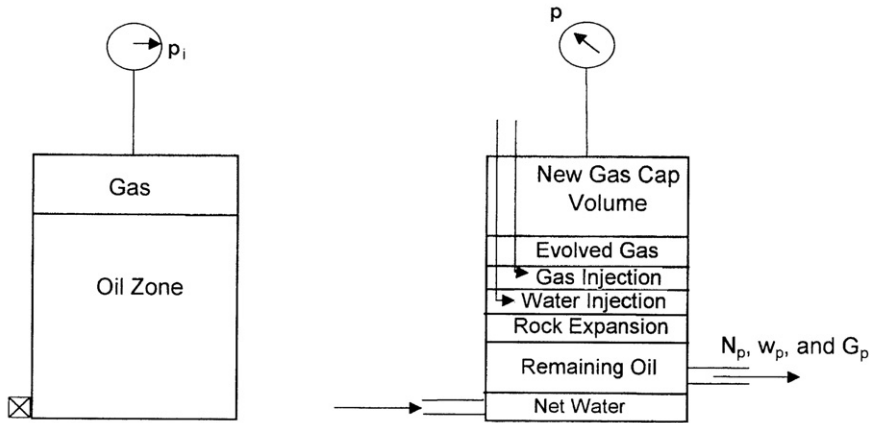


Figure 11-14. Tank-model concept.

$$\begin{aligned}
 &+ \\
 &\text{Pore volume occupied by the gas in the gas cap at } p \\
 &+ \\
 &\text{Pore volume occupied by the evolved solution gas at } p \\
 &+ \\
 &\text{Pore volume occupied by the net water influx at } p \\
 &+ \\
 &\text{Change in pore volume due to connate-water expansion and pore} \\
 &\text{volume reduction due to rock expansion} \\
 &+ \\
 &\text{Pore volume occupied by the injected gas at } p \\
 &+ \\
 &\text{Pore volume occupied by the injected water at } p
 \end{aligned} \tag{11-2}$$

The above nine terms composing the MBE can be separately determined from the hydrocarbon PVT and rock properties, as follows:

Pore Volume Occupied by the Oil Initially in Place

$$\text{Volume occupied by initial oil-in-place} = N B_{oi} \tag{11-3}$$

where N = oil initially in place, STB

B_{oi} = oil formation volume factor at initial reservoir pressure p_i ,
bbl/STB

Pore Volume Occupied by the Gas in the Gas Cap

$$\text{Volume of gas cap} = m N B_{oi} \quad (11-4)$$

where m is a dimensionless parameter and defined as the ratio of gas-cap volume to the oil zone volume.

Pore Volume Occupied by the Remaining Oil

$$\text{Volume of the remaining oil} = (N - N_p) B_o \quad (11-5)$$

where N_p = cumulative oil production, STB

B_o = oil formation volume factor at reservoir pressure p , bbl/STB

Pore Volume Occupied by the Gas Cap at Reservoir Pressure p

As the reservoir pressure drops to a new level p , the gas in the gas cap expands and occupies a larger volume. Assuming no gas is produced from the gas cap during the pressure decline, the new volume of the gas cap can be determined as:

$$\text{Volume of the gas cap at } p = \left[\frac{m N B_{oi}}{B_{gi}} \right] B_g \quad (11-6)$$

where B_{gi} = gas formation volume factor at initial reservoir pressure, bbl/scf

B_g = current gas formation volume factor, bbl/scf

Pore Volume Occupied by the Evolved Solution Gas

This volumetric term can be determined by applying the following material balance on the solution gas:

$$\begin{aligned} \left[\begin{array}{c} \text{volume of the evolved} \\ \text{solution gas} \end{array} \right] &= \left[\begin{array}{c} \text{volume of gas initially} \\ \text{in solution} \end{array} \right] \\ &\quad - \left[\begin{array}{c} \text{volume of gas} \\ \text{produced} \end{array} \right] \\ &\quad - \left[\begin{array}{c} \text{volume of gas} \\ \text{remaining in solution} \end{array} \right] \end{aligned}$$

or

$$\left[\begin{array}{c} \text{volume of the evolved} \\ \text{solution gas} \end{array} \right] = [N R_{si} - N_p R_p - (N - N_p) R_s] B_g \quad (11-7)$$

where N_p = cumulative oil produced, STB

R_p = net cumulative produced gas-oil ratio, scf/STB

R_s = current gas solubility factor, scf/STB

B_g = current gas formation volume factor, bbl/scf

R_{si} = gas solubility at initial reservoir pressure, scf/STB

Pore Volume Occupied by the Net Water Influx

$$\text{net water influx} = W_e - W_p B_w \quad (11-8)$$

where W_e = cumulative water influx, bbl

W_p = cumulative water produced, STB

B_w = water formation volume factor, bbl/STB

Change in Pore Volume Due to Initial Water and Rock Expansion

The component describing the reduction in the hydrocarbon pore volume due to the expansion of initial (connate) water and the reservoir rock cannot be neglected for an undersaturated-oil reservoir. The water compressibility c_w and rock compressibility c_f are generally of the same order of magnitude as the compressibility of the oil. The effect of these two components, however, can be generally neglected for the gas-cap-drive reservoir or when the reservoir pressure drops below the bubble-point pressure.

The compressibility coefficient c , which describes the changes in the volume (expansion) of the fluid or material with changing pressure, is given by:

$$c = \frac{-1}{V} \frac{\partial V}{\partial p}$$

or

$$\Delta V = V c \Delta p$$

where ΔV represents the net changes or expansion of the material as a result of changes in the pressure. Therefore, the reduction in the pore volume due to the expansion of the connate-water in the oil zone and the gas cap is given by:

$$\text{Connate-water expansion} = [(\text{pore volume}) S_{wi}] c_w \Delta p$$

Substituting for the pore volume (P.V) with Equation 11-1 gives:

$$\text{Expansion of connate water} = \frac{N B_{oi} (1 + m)}{1 - S_{wi}} S_{wi} c_w \Delta p \quad (11-9)$$

where Δp = change in reservoir pressure, $(p_i - p)$

c_w = water compressibility coefficient, psi^{-1}

m = ratio of the volume of the gas-cap gas to the reservoir oil volume, bbl/bbl

Similarly, the reduction in the pore volume due to the expansion of the reservoir rock is given by:

$$\text{Change in pore volume} = \frac{N B_{oi} (1 + m)}{1 - S_{wi}} c_f \Delta p \quad (11-10)$$

Combining the expansions of the connate-water and formation as represented by Equations 11-9 and 11-10 gives:

Total changes in the pore volume

$$= N B_{oi} (1 + m) \left(\frac{S_{wi} c_w + c_f}{1 - S_{wi}} \right) \Delta p \quad (11-11)$$

Pore Volume Occupied by the Injection Gas and Water

Assuming that G_{inj} volumes of gas and W_{inj} volumes of water have been injected for pressure maintenance, the total pore volume occupied by the two injected fluids is given by:

$$\text{Total volume} = G_{inj} B_{ginj} + W_{inj} B_w \quad (11-12)$$

where G_{inj} = cumulative gas injected, scf
 B_{ginj} = injected gas formation volume factor, bbl/scf
 W_{inj} = cumulative water injected, STB
 B_w = water formation volume factor, bbl/STB

Combining Equations 11-3 through 11-12 with Equation 11-2 and rearranging gives:

$$N = \frac{N_p B_o + (G_p - N_p R_s) B_g - (W_e - W_p B_w) - G_{inj} B_{ginj} - W_{inj} B_w}{(B_o - B_{oi}) + (R_{si} - R_s) B_g + m B_{oi} \left[\frac{B_g}{B_{gi}} - 1 \right] + B_{oi} (1 + m) \left[\frac{S_{wi} c_w + c_f}{1 - S_{wi}} \right] \Delta p} \quad (11-13)$$

where N = initial oil-in-place, STB
 G_p = cumulative gas produced, scf
 N_p = cumulative oil produced, STB
 R_{si} = gas solubility at initial pressure, scf/STB
 m = ratio of gas-cap gas volume to oil volume, bbl/bbl
 B_{gi} = gas formation volume factor at p_i , bbl/scf
 B_{ginj} = gas formation volume factor of the injected gas, bbl/scf

The cumulative gas produced G_p can be expressed in terms of the cumulative gas-oil ratio R_p and cumulative oil produced N_p by:

$$G_p = R_p N_p \quad (11-14)$$

Combining Equation 11-14 with Equation 11-13 gives:

$$N = \frac{N_p [B_o + (R_p - R_s) B_g] - (W_e - W_p B_w) - G_{inj} B_{ginj} - W_{inj} B_w}{(B_o - B_{oi}) + (R_{si} - R_s) B_g + m B_{oi} \left[\frac{B_g}{B_{gi}} - 1 \right] + B_{oi} (1 + m) \left[\frac{S_{wi} c_w + c_f}{1 - S_{wi}} \right] \Delta p} \quad (11-15)$$

The above relationship is referred to as the **material balance equation** (MBE). A more convenient form of the MBE can be determined by introducing the concept of the total (two-phase) formation volume factor B_t into the equation. This oil PVT property is defined as:

$$B_t = B_o + (R_{si} - R_s) B_g \quad (11-16)$$

Introducing B_t into Equation 11-15 and assuming, for the sake of simplicity, no water or gas injection gives:

$$N = \frac{N_p [B_t + (R_p - R_{si})B_g] - (W_e - W_p B_w)}{(B_t - B_{ti}) + m B_{ti} \left[\frac{B_g}{B_{gi}} - 1 \right] + B_{ti} (1 + m) \left[\frac{S_{wi} c_w + c_f}{1 - S_{wi}} \right] \Delta p} \quad (11-17)$$

where S_{wi} = initial water saturation

R_p = cumulative produced gas-oil ratio, scf/STB

Δp = change in the volumetric average reservoir pressure, psi

In a combination-drive reservoir where all the driving mechanisms are simultaneously present, it is of practical interest to determine the relative magnitude of each of the driving mechanisms and its contribution to the production.

Rearranging Equation 11-17 gives:

$$\begin{aligned} \frac{N(B_t - B_{ti})}{A} + \frac{NmB_{ti}(B_g - B_{gi})/B_{gi}}{A} + \frac{W_e - W_p B_w}{A} \\ + \frac{NB_{oi}(1 + m) \left[\frac{c_w S_{wi} + c_f}{1 - S_{wi}} \right] (p_i - p)}{A} = 1 \end{aligned} \quad (11-18)$$

with the parameter A as defined by:

$$A = N_p [B_t + (R_p - R_{si}) B_g] \quad (11-19)$$

Equation 11-18 can be abbreviated and expressed as:

$$DDI + SDI + WDI + EDI = 1.0 \quad (11-20)$$

where DDI = depletion-drive index

SDI = segregation (gas-cap)-drive index

WDI = water-drive index

EDI = expansion (rock and liquid)-drive index

The four terms of the left-hand side of Equation 11-20 represent the major primary driving mechanisms by which oil may be recovered from oil reservoirs. As presented earlier in this chapter, these driving forces are:

a. Depletion Drive. Depletion drive is the oil recovery mechanism wherein the production of the oil from its reservoir rock is achieved by the expansion of the original oil volume with all its original dissolved gas. This driving mechanism is represented mathematically by the first term of Equation 11-18 or:

$$DDI = N (B_t - B_{ti})/A \quad (11-21)$$

where DDI is termed the **depletion-drive index**.

b. Segregation Drive. Segregation drive (gas-cap drive) is the mechanism wherein the displacement of oil from the formation is accomplished by the expansion of the original free gas cap. This driving force is described by the second term of Equation 11-18, or:

$$SDI = [N m B_{ti} (B_g - B_{gi})/B_{gi}]/A \quad (11-22)$$

where SDI is termed the **segregation-drive index**.

c. Water Drive. Water drive is the mechanism wherein the displacement of the oil is accomplished by the net encroachment of water into the oil zone. This mechanism is represented by the third term of Equation 11-18 or:

$$WDI = (W_e - W_p B_w)/A \quad (11-23)$$

where WDI is termed the **water-drive index**.

d. Expansion Drive. For undersaturated-oil reservoirs with no water influx, the principal source of energy is a result of the rock and fluid expansion. Where all the other three driving mechanisms are contributing to the production of oil and gas from the reservoir, the contribution of the rock and fluid expansion to the oil recovery is too small and essentially negligible and can be ignored.

Cole (1969) pointed out that since the sum of the driving indexes is equal to one, it follows that if the magnitude of one of the index terms is reduced, then one or both of the remaining terms must be correspondingly increased. An effective water drive will usually result in maximum recovery from the reservoir. Therefore, if possible, the reservoir should be operated to yield a maximum water-drive index and minimum values for the depletion-drive index and the gas-cap-drive index. Maximum advantage should be taken of the most efficient drive available, and where the water drive is too weak to provide an effective displacing force, it may be possible to utilize the displacing energy of the gas cap. In any event, the depletion-drive index should be maintained as low as possible at all times, as this is normally the most inefficient driving force available.

Equation 11-20 can be solved at any time to determine the magnitude of the various driving indexes. The forces displacing the oil and gas from the reservoir are subject to change from time to time and for this reason Equation 11-20 should be solved periodically to determine whether there has been any change in the driving indexes. Changes in fluid withdrawal rates are primarily responsible for changes in the driving indexes. For example, reducing the oil-producing rate could result in an increased water-drive index and a correspondingly reduced depletion-drive index in a reservoir containing a weak water drive. Also, by shutting in wells producing large quantities of water, the water-drive index could be increased, as the net water influx (gross water influx minus water production) is the important factor.

When the reservoir has a very weak water drive but a fairly large gas cap, the most efficient reservoir producing mechanism may be the gas cap, in which case a large gas-cap-drive index is desirable. Theoretically, recovery by gas-cap drive is independent of producing rate, as the gas is readily expansible. Low vertical permeability could limit the rate of expansion of the gas cap, in which case the gas-cap-drive index would be rate sensitive. Also, gas coning into producing wells will reduce the effectiveness of the gas-cap expansion due to the production of free gas. Gas coning is usually a rate sensitive phenomenon; the higher the producing rates, the greater the amount of coning.

An important factor in determining the effectiveness of a gas-cap drive is the degree of conservation of the gas-cap gas. As a practical matter, it will often be impossible, because of royalty owners or lease agreements, to completely eliminate gas-cap gas production. Where free gas is being produced, the gas-cap-drive index can often be markedly increased by

shutting in high gas-oil ratio wells and, if possible, transferring their allowables to other low gas-oil ratio wells.

Figure 11-15 shows a set of plots that represent various driving indexes for a combination-drive reservoir. At point A, some of the structurally low wells are reworked to reduce water production. This resulted in an effective increase in the water-drive index. At point B, workover operations are complete; water-, gas-, and oil-producing rates are relatively stable; and the driving indexes show no change. At point C, some of the wells which have been producing relatively large, but constant, volumes of water are shut-in, which results in an increase in the water-drive index. At the same time, some of the upstructure, high gas-oil ratio wells have been shut-in and their allowables transferred to wells lower on the structure producing with normal gas-oil ratios. At point D, gas is being returned to the reservoir, and the gas-cap-drive index is exhibiting a decided increase.

The water-drive index is relatively constant, although it is decreasing somewhat, and the depletion-drive index is showing a marked decline. This is indicative of a more efficient reservoir operation, and, if the depletion-drive index can be reduced to zero, relatively good recovery can be expected from the reservoir. Of course, to achieve a zero-depletion-drive index would require the complete maintenance of reservoir pressure, which is often difficult to accomplish. It can be noted from Figure 11-15 that the sum of the various indexes of drive is always equal to one.

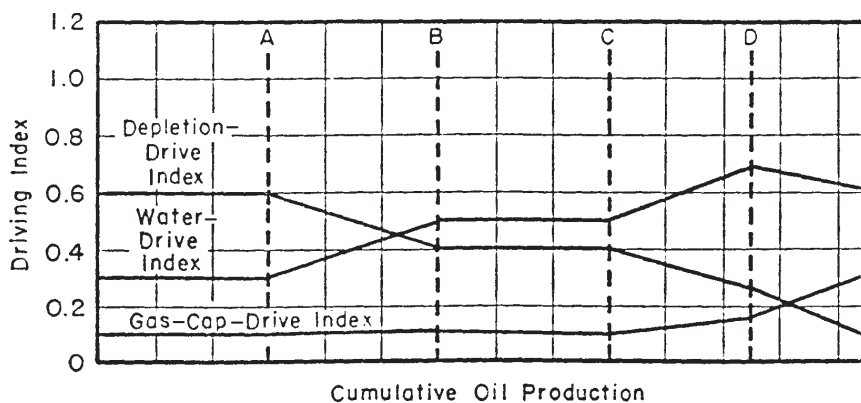


Figure 11-15. Driving indexes in a combination-drive reservoir. (After Clark, N. J., *Elements of Petroleum Reservoirs*, SPE, 1969.)

Example 11-1

A combination-drive reservoir contains 10 MMSTB of oil initially in place. The ratio of the original gas-cap volume to the original oil volume, i.e., m , is estimated as 0.25. The initial reservoir pressure is 3,000 psia at 150°F. The reservoir produced 1 MMSTB of oil, 1,100 MMscf of 0.8 specific gravity gas, and 50,000 STB of water by the time the reservoir pressure dropped to 2,800 psi. The following PVT is available:

	3000 psi	2800 psi
B_o , bbl/STB	1.58	1.48
R_s , scf/STB	1040	850
B_g , bbl/scf	0.00080	0.00092
B_t , bbl/STB	1.58	1.655
B_w , bbl/STB	1.000	1.000

The following data are also available:

$$S_{wi} = 0.20 \quad c_w = 1.5 \times 10^{-6} \text{ psi}^{-1} \quad c_f = 1 \times 10^{-6} \text{ psi}^{-1}$$

Calculate:

- Cumulative water influx
- Net water influx
- Primary driving indexes at 2,800 psi

Solution

Because the reservoir contains a gas cap, the rock and fluid expansion can be neglected, i.e., set c_f and $c_w = 0$. For illustration purposes, however, the rock and fluid expansion term will be included in the calculations.

Part A. Cumulative water influx

Step 1. Calculate cumulative gas-oil ratio R_p :

$$R_p = \frac{1100 \times 10^6}{1 \times 10^6} = 1100 \text{ scf/STB}$$

Step 2. Arrange Equation 11-17 to solve for W_e :

$$\begin{aligned}
 W_e &= N_p [B_t + (R_p - R_{si}) B_g] - N \left[(B_g - B_{ti}) + m B_{ti} \left(\frac{B_g}{B_{gi}} - 1 \right) \right. \\
 &\quad \left. + B_{ti} (1 + m) \left(\frac{S_{wi} c_w + c_f}{1 - S_{wi}} \right) \Delta p \right] + W_p B_{wp} \\
 W_e &= 10^6 [1.655 + (1100 - 1040) 0.00092] \\
 &\quad - 10^7 \left[(1.655 - 1.58) + 0.25 (1.58) \left(\frac{0.00092}{0.00080} - 1 \right) \right. \\
 &\quad \left. + 1.58 (1 + 0.25) \left(\frac{0.2 (1.5 \times 10^{-6})}{1 - 0.2} \right) (3000 - 2800) \right] + 50,000 \\
 &= 411,281 \text{ bbl}
 \end{aligned}$$

Neglecting the rock and fluid expansion term, the cumulative water influx is 417,700 bbl.

Part B. Net water influx

$$\text{Net water influx} = W_e - W_p B_w = 411,281 - 50,000 = 361,281 \text{ bbl}$$

Part C. Primary recovery indexes

Step 1. Calculate the parameter A by using Equation 11-19:

$$A = 10^6 [1.655 + (1100 - 1040) 0.00092] = 1,710,000$$

Step 2. Calculate DDI, SDI, and WDI by applying Equations 11-21 through 11-23, respectively:

$$\text{DDI} = \frac{10 \times 10^6 (1.655 - 1.58)}{1,710,000} = 0.4385$$

$$\text{SDI} = \frac{10 \times 10^6 (0.25) (1.58) (0.00092 - 0.0008) / 0.0008}{1,710,000} = 0.3465$$

$$\text{WDI} = \frac{411,281 - 50,000}{1,710,000} = 0.2112$$

$$\text{EDI} = 1 - 0.4385 - 0.3465 - 0.2112 = 0.0038$$

where EDI is termed the **expansion-drive index**.

These calculations show that the 43.85% of the recovery was obtained by depletion drive, 34.65% by gas-cap drive, 21.12% by water drive, and only 0.38% by connate-water and rock expansion. The results suggest that the expansion-drive index (EDI) term can be neglected in the presence of a gas cap or when the reservoir pressure drops below the bubble-point pressure. In high pore volume compressibility reservoirs, such as chalks and unconsolidated sands, however, the energy contribution of the rock and water expansion cannot be ignored even at high gas saturations.

Example 11-2

The Big Butte field is a combination-drive reservoir. The current reservoir pressure is estimated at 2,500 psi. The reservoir production data and PVT information are given below:

	Initial Reservoir Condition	Current Reservoir Condition
p, psi	3000	2500
B _o , bbl/STB	1.35	1.33
R _s , scf/STB	600	500
N _p , MMSTB	0	5
G _p , MMMscf		5.5
B _w , bbl/STB	1.00	1.00
W _e , MMbbl	0	3
W _p , MMbbl	0	0.2
B _g , bbl/scf	0.0011	0.0015
c _f , c _w	0	0

The following additional information is available:

Volume of bulk oil zone = 100,000 ac-ft

Volume of bulk gas zone = 20,000 ac-ft

Calculate the initial oil-in-place.

Solution

Step 1. Assuming the same porosity and connate-water for the oil and gas zones, calculate m :

$$m = \frac{20,000}{100,000} = 0.2$$

Step 2. Calculate the cumulative gas-oil ratio R_p :

$$R_p = \frac{5.5 \times 10^9}{5 \times 10^6} = 1100 \text{ scf/STB}$$

Step 3. Solve for the initial oil-in-place by applying Equation 11-15:

$$\begin{aligned} N &= \frac{5 \times 10^6 [1.33 + (1100 - 500)0.0015] - (3 \times 10^6 - 0.2 \times 10^6)}{(1.35 - 1.33) + (600 - 500)0.0015 + (0.2)(1.35) \left[\frac{0.0015}{0.0011} - 1 \right]} \\ &= 31.14 \text{ MMSTB} \end{aligned}$$

Basic Assumptions in the MBE

The material balance equation calculation is based on changes in reservoir conditions over discrete periods of time during the production history. The calculation is most vulnerable to many of its underlying assumptions early in the depletion sequence when fluid movements are limited and pressure changes are small. Uneven depletion and partial reservoir development compound the accuracy problem.

The basic assumptions in the material balance equation (MBE) are as follows:

- **Constant temperature.** Pressure-volume changes in the reservoir are assumed to occur without any temperature changes. If any temperature changes occur, they are usually sufficiently small to be ignored without significant error.
- **Pressure equilibrium.** All parts of the reservoir have the same pressure, and fluid properties are therefore constant throughout. Minor variations in the vicinity of the well bores may usually be ignored. Substantial pressure variation across the reservoir may cause excessive calculation error.

It is assumed that the PVT samples or data sets represent the actual fluid compositions and that reliable and representative laboratory procedures have been used. Notably, the vast majority of material balances assume that differential depletion data represent reservoir flow and that separator flash data may be used to correct for the wellbore transition to surface conditions. Such “black oil” PVT treatments relate volume changes to temperature and pressure only. They lose validity in cases of volatile oil or gas condensate reservoirs where compositions are also important. Special laboratory procedures may be used to improve PVT data for volatile fluid situations.

Constant reservoir volume. Reservoir volume is assumed to be constant except for those conditions of rock and water expansion or water influx that are specifically considered in the equation. The formation is considered to be sufficiently competent that no significant volume change will occur through movement or reworking of the formation due to overburden pressure as the internal reservoir pressure is reduced. The constant volume assumption is also related to an area of interest to which the equation is applied. If the focus is on some part of a reservoir system, except for specific exterior flow terms it is assumed that the particular portion is encased in no-flow boundaries.

Reliable production data. All production data should be recorded with respect to the same time period. If possible, gas-cap- and solution-gas production records should be maintained separately.

Gas and oil gravity measurements should be recorded in conjunction with the fluid volume data. Some reservoirs require a more detailed analysis and that the material balance be solved for volumetric segments. The produced fluid gravities will aid in the selection of the volumetric segments and also in the averaging of fluid properties. There are essentially three types of production data that must be recorded in order to use the MBE in performing reliable reservoir calculations. These are:

- **Oil-production data**, even for non-interest properties, which can usually be obtained from various sources and are usually fairly reliable.
- **Gas-production data**, which are becoming more available and reliable as the market value of this commodity increases; unfortunately, these data will often be more questionable where gas is flared.
- **The water-production term**, which need represent only the net withdrawals of water; therefore, where subsurface disposal of produced brine is to the same source formation, most of the error due to poor data will be eliminated.

A source of error is often introduced in the MBE calculations when determining the average reservoir pressure and the associated problem of correctly weighting or averaging the individual well pressures. An example of such a problem is when the producing formations are composed of two or more zones of different permeabilities. In this case, the pressures are generally higher in the zone of low permeability, and because the measured pressures are nearer to those in high-permeability zones, the measured static pressures tend to be lower and the reservoir behaves as if it contained less oil. Schilthuis (1941) explained this phenomenon by referring to the oil in the more permeable zones as **active oil** and by observing that the *calculated active oil usually increases with time* because the oil and gas in low-permeability zones slowly expand to offset the pressure decline. This is also true for fields that are not fully developed, because the average pressure can be that of only the developed portion, whereas the pressure is higher in the undeveloped portions.

Craft et al. (1991) pointed out that the effect of pressure errors on the calculated values of initial oil and water influx depends on the size of the errors in relation to the reservoir pressure decline. Notice that the pressure enters the MBE mainly when determining the PVT differences in terms of the following:

- $(B_o - B_{oi})$
- $(B_g - B_{gi})$
- $(R_{si} - R_s)$

Because water influx and gas cap expansion tend to offset pressure decline, the pressure errors are more serious than for the undersaturated reservoirs. In the case of very active water drives or gas caps that are large compared with the oil zone, the MBE usually produces considerable errors when determining the initial oil-in-place because of the very small pressure decline.

Dake (1994) points out that there are two “necessary” conditions that must be satisfied for a meaningful application of the MBE to a reservoir; these are as follows:

1. There should be adequate data collection in terms of production pressure and PVT, both in frequency and quality for proper use of the MBE.
2. It must be possible to define an average reservoir pressure trend as a function of time or production for the field.

Establishing an average pressure-decline trend can be possible even if there are large pressure differences across the field under normal conditions. Averaging **individual well pressure** declines can possibly be used to determine a uniform trend in the entire reservoir. The concept of average well pressure and its use in determining the reservoir volumetric average pressure was introduced in Chapter 6, as illustrated by Figure 6-24. This illustration shows that if $(\bar{P})_j$ and V_j represent the pressure and volume drained by the j th well, the volumetric average pressure of the entire reservoir can be estimated from:

$$\bar{p}_r = \frac{\sum_j (\bar{p} V)_j}{\sum_j V_j}$$

in which

V_j = pore volume of the j th well drainage volume

$(\bar{P})_j$ = Volumetric average pressure *within the j th drainage volume*

In practice, the V_i values are difficult to determine and, therefore, it is common to use individual wells' flow rates, q_i , in determining the average reservoir pressure from individual wells' average drainage pressure. From the definition of the isothermal compressibility coefficient:

$$c = \frac{1}{V} \frac{\partial V}{\partial P}$$

Differentiating this equation with time gives:

$$\frac{\partial p}{\partial t} = \frac{1}{c V} \frac{\partial V}{\partial t}$$

or:

$$\frac{\partial p}{\partial t} = \frac{1}{c V} (q)$$

The last expression suggests that for a reasonably constant c at the time of measurement,

$$V \propto \frac{q}{\partial p / \partial t}$$

Since the flow rates are measured on a routing basis throughout the lifetime of the field, the average reservoir pressure can be alternatively

expressed in terms of the individual wells' average drainage-pressure-decline rates and fluid-flow rates by:

$$\bar{p}_r = \frac{\sum_j [(\bar{p}q)_j / (\partial \bar{p} / \partial t)_j]}{\sum_j [q_j / (\partial \bar{p} / \partial t)_j]}$$

However, since the MBE is usually applied at regular intervals of 3–6 months (i.e., $\Delta t = 3\text{--}6$ months), throughout the life of the field, the average field pressure can be expressed in terms of the incremental net change in underground fluid withdrawal, $\Delta(F)$, as follows:

$$\bar{p}_r = \frac{\sum_j \frac{\bar{p}_j \Delta(F)_j}{\Delta \bar{p}_j}}{\sum_j \frac{\Delta(F)_j}{\Delta \bar{p}_j}}$$

where the total underground fluid withdrawal at times t and $t + \Delta t$ are given by:

$$F_t = \int_0^t [Q_o B_o + Q_w B_w + (Q_g - Q_o R_s - Q_w R_{sw}) B_g] dt$$

$$F_{t+\Delta t} = \int_0^{t+\Delta t} [Q_o B_o + Q_w B_w + (Q_g - Q_o R_s - Q_w R_{sw}) B_g] dt$$

with:

$$\Delta(F) = F_{t+\Delta t} - F_t$$

where R_s = gas solubility, scf/STB

R_{sw} = gas solubility in the water, scf/STB

B_g = gas formation volume factor, bbl/scf

Q_o = oil flow rate, STB/day

q_o = oil flow rate, bbl/day

Q_w = water flow rate, STB/day

q_w = water flow rate, bbl/day

Q_g = gas flow rate, scf/day

For a volumetric reservoir with total fluid production and initial reservoir pressure as the only available data, the average pressure can be *roughly approximated* by using the following expression:

$$\bar{p}_r = p_i - \left[\frac{5.371 \times 10^{-6} F_t}{c_t (Ah\phi)} \right]$$

with the total fluid production, F_t , as defined above by:

$$F_t = \int_0^t [Q_o B_o + Q_w B_w + (Q_g - Q_o R_s - Q_w R_{sw}) B_g] dt$$

where A = well or reservoir drainage area, acres

h = thickness, ft

c_t = total compressibility coefficient, psi^{-1}

ϕ = porosity, fraction

p_i = initial reservoir pressure, psi

The last expression can be employed in an incremental manner, from time t to $t + \Delta t$, by:

$$(\bar{p}_r)_{t+\Delta t} = (\bar{p}_r)_t - \left[\frac{5.371 \times 10^{-6} \Delta F}{c_t (Ah\phi)} \right]$$

with:

$$\Delta(F) = F_{t+\Delta t} - F_t$$

The MBE as an Equation of a Straight Line

An insight into the general MBE, i.e., Equation 11-15, may be gained by considering the physical significance of the following groups of terms of which it is comprised:

- $N_p [B_o + (R_p - R_s) B_g]$ Represents the reservoir volume of cumulative oil and gas produced.
- $[W_e - W_p B_w]$ Refers to the net water influx that is retained in the reservoir.
- $[G_{inj} B_{ginj} + W_{inj} B_w]$ This pressure maintenance term represents cumulative fluid injection in the reservoir.

- $[m B_{oi} (B_g/B_{gi} - 1)]$ Represents the net expansion of the gas cap that occurs with the production of N_p stock-tank barrels of oil (as expressed in bbl/STB of original oil-in-place).

There are essentially three unknowns in Equation 11-15:

- a. The original oil-in-place N
- b. The cumulative water influx W_e
- c. The original size of the gas cap as compared to the oil zone size m

In developing a methodology for determining the above three unknowns, Havlena and Odeh (1963) expressed Equation 11-15 in the following form:

$$N_p [B_o + (R_p - R_s) B_g] + W_p B_w = N[(B_o - B_{oi}) + (R_{si} - R_s) B_g] + m N B_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right) + N (1 + m) B_{oi} \left[\frac{c_w S_{wi} + c_f}{1 + S_{wi}} \right] \Delta p + W_e + W_{inj} B_w + G_{inj} B_{ginj} \quad (11-24)$$

Havlena and Odeh further expressed Equation 11-24 in a more condensed form as:

$$F = N [E_o + m E_g + E_{f,w}] + (W_e + W_{inj} B_w + G_{inj} B_{ginj})$$

Assuming, for the purpose of simplicity, that no pressure maintenance by gas or water injection is being considered, the above relationship can be further simplified and written as:

$$F = N [E_o + m E_g + E_{f,w}] + W_e \quad (11-25)$$

in which the terms F , E_o , E_g , and $E_{f,w}$ are defined by the following relationships:

- F represents the underground withdrawal and is given by:

$$F = N_p [B_o + (R_p - R_s) B_g] + W_p B_w \quad (11-26)$$

In terms of the two-phase formation volume factor B_t , the underground withdrawal F can be written as:

$$F = N_p [B_t + (R_p - R_{si}) B_g] + W_p B_w \quad (11-27)$$

- E_o describes the expansion of oil and its originally dissolved gas and is expressed in terms of the oil formation volume factor as:

$$E_o = (B_o - B_{oi}) + (R_{si} - R_s) B_g \quad (11-28)$$

Or equivalently, in terms of B_t :

$$E_o = B_t - B_{ti} \quad (11-29)$$

- E_g is the term describing the expansion of the gas-cap gas and is defined by the following expression:

$$E_g = B_{oi} [(B_g/B_{gi}) - 1] \quad (11-30)$$

In terms of the two-phase formation volume factor B_t , essentially $B_{ti} = B_{oi}$ or:

$$E_g = B_{ti} [(B_g/B_{gi}) - 1]$$

- $E_{f,w}$ represents the expansion of the initial water and the reduction in the pore volume and is given by:

$$E_{f,w} = (1 + m)B_{oi} \left[\frac{c_w S_{wi} + c_f}{1 - S_{wi}} \right] \Delta p \quad (11-31)$$

Havlena and Odeh examined several cases of varying reservoir types with Equation 11-25 and pointed out that the relationship can be rearranged into the form of a straight line. For example, in the case of a reservoir which has no initial gas cap (i.e., $m = 0$) or water influx (i.e., $W_e = 0$), and negligible formation and water compressibilities (i.e., c_f and $c_w = 0$), Equation 11-25 reduces to:

$$F = N E_o$$

The above expression suggests that a plot of the parameter F as a function of the oil expansion parameter E_o would yield a straight line with a slope N and intercept equal to zero.

The Straight-Line Solution Method to the MBE

The straight-line solution method requires the plotting of a variable group versus another variable group, with the variable group selection depending on the mechanism of production under which the reservoir is producing. The most important aspect of this method of solution is that it

attaches significance to the sequence of the plotted points, the direction in which they plot, and to the shape of the resulting plot.

The significance of the straight-line approach is that the sequence of plotting is important and if the plotted data deviate from this straight line, there is some reason for it. This significant observation will provide the engineer with valuable information that can be used in determining the following unknowns:

- Initial oil-in-place N
- Size of the gas cap m
- Water influx W_e
- Driving mechanism

The applications of the straight-line form of the MBE in solving reservoir engineering problems are presented next to illustrate the usefulness of this particular form. Six cases of applications are presented:

1. Case 1: Determination of N in volumetric undersaturated reservoirs
2. Case 2: Determination of N in volumetric saturated reservoirs
3. Case 3: Determination of N and m in gas-cap-drive reservoirs
4. Case 4: Determination of N and W_e in water-drive reservoirs
5. Case 5: Determination of N , m , and W_e in combination-drive reservoirs
6. Case 6: Determination of average reservoir pressure, $\bar{p}$

The remainder of this chapter is devoted to illustrations of the use of the straight-line solution method in determining N , m , and W_e for different reservoir mechanisms.

Case 1. Volumetric Undersaturated-Oil Reservoirs

Assuming no water or gas injection, the linear form of the MBE as expressed by Equation 11-25 can be written as:

$$F = N [E_o + m E_g + E_{f,w}] + W_e \quad (11-32)$$

Several terms in the above relationship may disappear when imposing the conditions associated with the assumed reservoir-driving mechanism. For a volumetric and undersaturated reservoir, the conditions associated with a driving mechanism are:

- $W_e = 0$, since the reservoir is volumetric
- $m = 0$, since the reservoir is undersaturated

- $R_s = R_{si} = R_p$, since all produced gas is dissolved in the oil

Applying the above conditions on Equation 11-32 gives:

$$F = N (E_o + E_{f,w}) \quad (11-33)$$

or

$$N = \frac{F}{E_o + E_{f,w}} \quad (11-34)$$

where N = initial oil-in-place, STB

$$F = N_p B_o + W_p B_w \quad (11-35)$$

$$E_o = B_o - B_{oi} \quad (11-36)$$

$$E_{f,w} = B_{oi} \left[\frac{c_w S_w + c_f}{1 - S_{wi}} \right] \Delta p \quad (11-37)$$

$$\Delta p = p_i - \bar{p}_r$$

p_i = initial reservoir pressure

$\bar{p}_r$ = volumetric average reservoir pressure

When a new field is discovered, one of the first tasks of the reservoir engineer is to determine if the reservoir can be classified as a volumetric reservoir, i.e., $W_e = 0$. The classical approach of addressing this problem is to assemble all the necessary data (i.e., production, pressure, and PVT) that are required to evaluate the right-hand side of Equation 11-36. The term $F/(E_o + E_{f,w})$ for each pressure and time observation is plotted versus cumulative production N_p or time, as shown in Figure 11-16. Dake (1994) suggests that such a plot can assume two various shapes, which are:

- All the calculated points of $F/(E_o + E_{f,w})$ lie on a horizontal straight line (see Line A in Figure 11-16). Line A in the plot implies that the reservoir can be classified as a volumetric reservoir. This defines a purely depletion-drive reservoir whose energy derives solely from the expansion of the rock, connate-water, and the oil. Furthermore, the ordinate value of the plateau determines the initial oil-in-place N .
- Alternately, the calculated values of the term $F/(E_o + E_{f,w})$ rise, as illustrated by the curves B and C, indicating that the reservoir has been

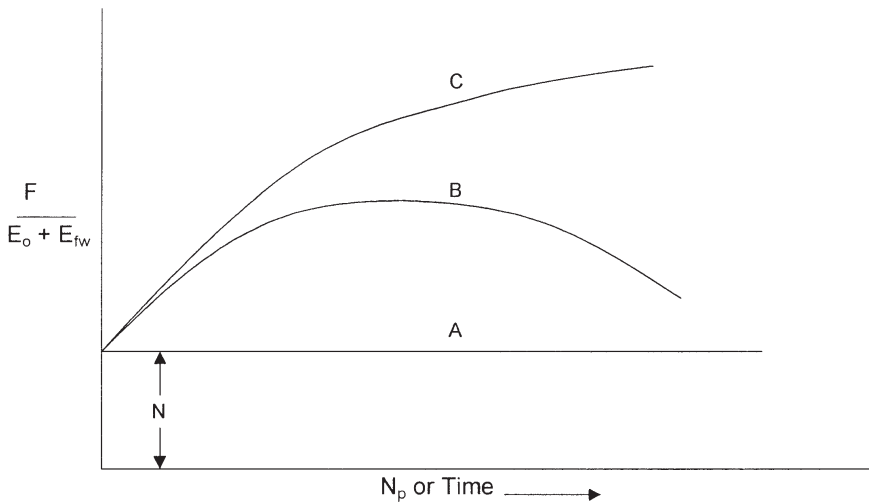


Figure 11-16. Classification of the reservoir.

energized by water influx, abnormal pore compaction, or a combination of these two. Curve C in Figure 11-16 might be for a strong water-drive field in which the aquifer is displacing an infinite acting behavior, whereas B represents an aquifer whose outer boundary has been felt and the aquifer is depleting in unison with the reservoir itself. The downward trend in points on curve B as time progresses denotes the diminishing degree of energizing by the aquifer. Dake (1994) points out that in water-drive reservoirs, the shape of the curve, i.e., $F/(E_o + E_{f,w})$ vs. time, is highly rate dependent. For instance, if the reservoir is producing at a higher rate than the water-influx rate, the calculated values of $F/(E_o + E_{f,w})$ will dip downward revealing a lack of energizing by the aquifer, whereas, if the rate is decreased, the reverse happens and the points are elevated.

Similarly, Equation 11-33 could be used to verify the characteristic of the reservoir-driving mechanism and to determine the initial oil-in-place. A plot of the underground withdrawal F versus the expansion term $(E_o + E_{f,w})$ should result in a straight line going through the origin with N being the slope. It should be noted that the origin is a “must” point; thus, one has a fixed point to guide the straight-line plot (as shown in Figure 11-17).

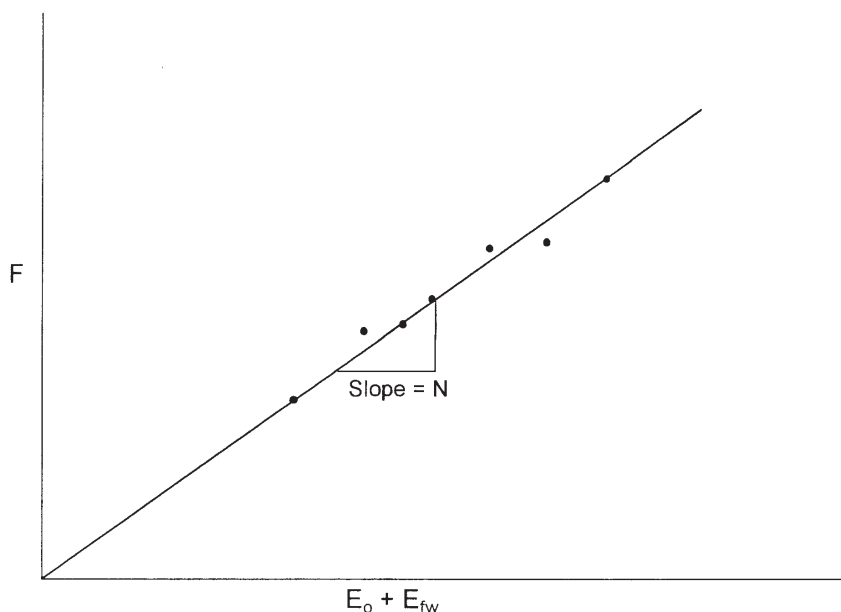


Figure 11-17. Underground withdrawal vs. $E_o + E_{fw}$.

This interpretation technique is useful in that, if the linear relationship is expected for the reservoir and yet the actual plot turns out to be non-linear, then this deviation can itself be diagnostic in determining the actual drive mechanisms in the reservoir.

A linear plot of the underground withdrawal F versus $(E_o + E_{f,w})$ indicates that the field is producing under volumetric performance, i.e., no water influx, and strictly by pressure depletion and fluid expansion. On the other hand, a nonlinear plot indicates that the reservoir should be characterized as a water-drive reservoir.

Example 11-3

The Virginia Hills Beaverhill Lake field is a volumetric undersaturated reservoir. Volumetric calculations indicate the reservoir contains 270.6 MMSTB of oil initially in place. The initial reservoir pressure is 3685 psi. The following additional data are available:

$$\begin{array}{lll} S_{wi} = 24\% & c_w = 3.62 \times 10^{-6} \text{ psi}^{-1} & c_f = 4.95 \times 10^{-6} \text{ psi}^{-1} \\ B_w = 1.0 \text{ bbl/STB} & p_b = 1500 \text{ psi} & \end{array}$$

The field production and PVT data are summarized below:

Volumetric Average Pressure	No. of Producing Wells	B _o bbl/STB	N _p MSTB	W _p MSTB
3685	1	1.3102	0	0
3680	2	1.3104	20.481	0
3676	2	1.3104	34.750	0
3667	3	1.3105	78.557	0
3664	4	1.3105	101.846	0
3640	19	1.3109	215.681	0
3605	25	1.3116	364.613	0
3567	36	1.3122	542.985	0.159
3515	48	1.3128	841.591	0.805
3448	59	1.3130	1273.530	2.579
3360	59	1.3150	1691.887	5.008
3275	61	1.3160	2127.077	6.500
3188	61	1.3170	2575.330	8.000

Calculate the initial oil-in-place by using the MBE and compare with the volumetric estimate of N.

Solution

Step 1. Calculate the initial water and rock expansion term $E_{f,w}$ from Equation 11-37:

$$E_{f,w} = 1.3102 \left[\frac{3.62 \times 10^{-6} (0.24) + 4.95 \times 10^{-6}}{1 - 0.24} \right] \Delta p$$

$$E_{f,w} = 10.0 \times 10^{-6} (3685 - \bar{p}_r)$$

Step 2. Construct the following table:

$\bar{p}_r$, psi	F, Mbbl Equation 10-35	E _o , bbl/STB Equation 10-36	Δp	E _{f,w}	E _o + E _{f,w}
3685	—	—	0	0	—
3680	26.84	0.0002	5	50×10^{-6}	0.00025
3676	45.54	0.0002	9	90×10^{-6}	0.00029
3667	102.95	0.0003	18	180×10^{-6}	0.00048
3664	133.47	0.0003	21	210×10^{-6}	0.00051
3640	282.74	0.0007	45	450×10^{-6}	0.00115

(table continued on next page)

$\bar{p}_r$, psi	F, Mbbl Equation 10-35	E_o , bbl/STB Equation 10-36	Δp	$E_{f,w}$	$E_o + E_{f,w}$
3605	478.23	0.0014	80	800×10^{-6}	0.00220
3567	712.66	0.0020	118	1180×10^{-6}	0.00318
3515	1105.65	0.0026	170	1700×10^{-6}	0.00430
3448	1674.72	0.0028	237	2370×10^{-6}	0.00517
3360	2229.84	0.0048	325	3250×10^{-6}	0.00805
3275	2805.73	0.0058	410	4100×10^{-6}	0.00990
3188	3399.71	0.0068	497	4970×10^{-6}	0.01170

Step 3. Plot the underground withdrawal term F against the expansion term ($E_o + E_{f,w}$) on a Cartesian scale, as shown in Figure 11-18.

Step 4. Draw the best straight line through the points and determine the slope of the line and the volume of the active initial oil-in-place as:

$$N = 257 \text{ MMSTB}$$

It should be noted that the value of the initial oil-in-place as determined from the MBE is referred to as the effective or active initial oil-in-place. This value is usually smaller than that of the volumetric estimate due to oil being trapped in undrained fault compartments or low-permeability regions of the reservoir.

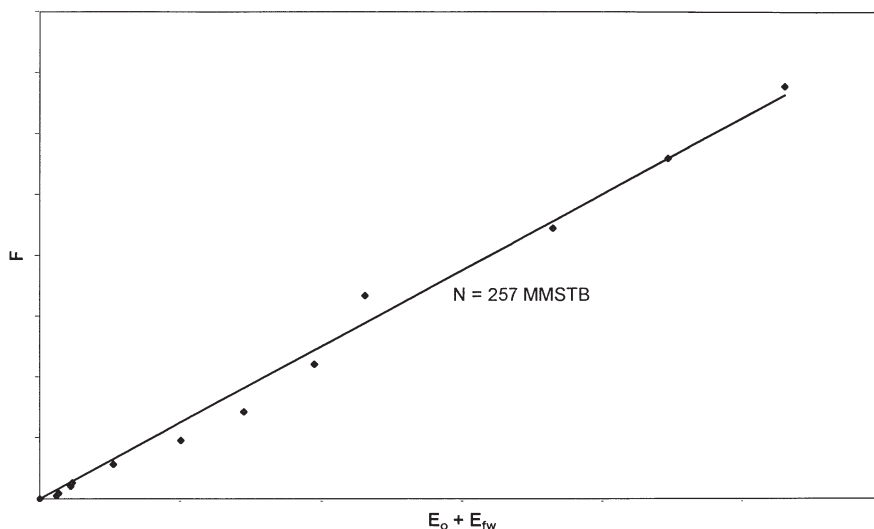


Figure 11-18. F vs. ($E_o + E_{fw}$) for Example 11-3.

Case 2. Volumetric Saturated-Oil Reservoirs

An oil reservoir that originally exists at its bubble-point pressure is referred to as a saturated-oil reservoir. The main driving mechanism in this type of reservoir results from the liberation and expansion of the solution gas as the pressure drops below the bubble-point pressure. The only unknown in a volumetric saturated-oil reservoir is the initial oil-in-place N . Assuming that the water and rock expansion term $E_{f,w}$ is negligible in comparison with the expansion of solution gas, Equation 11-32 can be simplified as:

$$F = N E_o \quad (11-38)$$

where the underground withdrawal F and the oil expansion E_o were defined previously by Equations 11-26 and 11-28 or Equations 11-27 and 11-29 to give:

$$F = N_p [B_t + (R_p - R_{si}) B_g] + W_p B_w$$

$$E_o = B_t - B_{ti}$$

Equation 11-38 indicates that a plot of the underground withdrawal F , evaluated by using the actual reservoir production data, as a function of the fluid expansion term E_o , should result in a straight line going through the origin with a slope of N .

The above interpretation technique is useful in that, if a simple linear relationship such as Equation 11-38 is expected for a reservoir and yet the actual plot turns out to be nonlinear, then this deviation can itself be diagnostic in determining the actual drive mechanisms in the reservoir. For instance, Equation 11-38 may turn out to be nonlinear because there is an unsuspected water influx into the reservoir helping to maintain the pressure.

It should be pointed out that, as the reservoir pressure continues to decline below the bubble-point and with the increasing volume of the liberated gas, it reaches the time where the saturation of the liberated gas exceeds the critical gas saturation. As a result, the gas will start to be produced in disproportionate quantities to the oil. At this stage of depletion, there is little that can be done to avert this situation during the primary production phase. As indicated earlier, the primary recovery from these types of reservoirs seldom exceeds 30%. However, under very favorable conditions, the oil and gas might separate, with the gas moving structurally updip in the reservoir; this might lead to preservation of the natur-

al energy of the reservoir with a consequent improvement in overall oil recovery. Water injection is traditionally used by the oil industry to maintain the pressure above the bubble-point pressure or alternatively to pressurize the reservoir to the bubble-point pressure. In such types of reservoirs, as the reservoir pressure drops below the bubble-point pressure, some volume of the liberated gas will remain in the reservoir as a free gas. This volume, expressed in scf, is given by Equation 11-7 as:

$$\left[\text{Volume of the free gas in scf} \right] = N R_{si} - (N - N_p) R_s - N_p R_p$$

However, the total volume of the liberated gas at any depletion pressure is given by

$$\left[\begin{array}{l} \text{Total volume of the} \\ \text{liberated gas, in scf} \end{array} \right] = N R_{si} - (N - N_p) R_s$$

Therefore, the fraction of the total solution gas that has been retained in the reservoir as a free gas, α_g , at any depletion stage is then given by:

$$\alpha_g = \frac{N R_{si} - (N - N_p) R_s - N_p R_p}{N R_{si} - (N - N_p) R_s} = 1 - \left[\frac{N_p R_p}{N R_{si} - (N - N_p) R_s} \right]$$

Alternatively, it can be expressed as a fraction of the total initial gas in solution, by

$$\alpha_{gi} = \frac{N R_{si} - (N - N_p) R_s - N_p R_p}{N R_{si}} = 1 - \left[\frac{(N - N_p) R_s + N_p R_p}{N R_{si}} \right]$$

The calculation of the changes in the fluid saturations with declining reservoir pressure is an integral part of using the MBE. The remaining volume of each phase can be determined by calculating the saturation of each phase as:

$$\text{Oil saturation: } S_o = \frac{\text{oil volume}}{\text{pore volume}}$$

$$\text{Water saturation: } S_w = \frac{\text{water volume}}{\text{pore volume}}$$

$$\text{Gas saturation: } S_g = \frac{\text{gas volume}}{\text{pore volume}}$$

and:

$$S_o + S_w + S_g = 1.0$$

If we consider a volumetric saturated-oil reservoir that contains N stock-tank barrels of oil at the initial reservoir pressure p_i (i.e., p_b), the initial oil saturation at the bubble-point pressure is given by:

$$S_{oi} = 1 - S_{wi}$$

From the definition of oil saturation:

$$\frac{\text{oil volume}}{\text{pore volume}} = \frac{N B_{oi}}{\text{pore volume}} = 1 - S_{wi}$$

or

$$\text{pore volume} = \frac{N B_{oi}}{1 - S_{wi}}$$

If the reservoir has produced N_p stock-tank barrels of oil, the remaining oil volume is given by:

$$\text{Remaining oil volume} = (N - N_p) B_o$$

This indicates that for a volumetric-type oil reservoir, the oil saturation at any depletion state below the bubble-point pressure can be represented by:

$$S_o = \frac{\text{oil volume}}{\text{pore volume}} = \frac{(N - N_p) B_o}{\left(\frac{N B_{oi}}{1 - S_{wi}} \right)}$$

Rearranging gives:

$$S_o = (1 - S_{wi}) \left(1 - \frac{N_p}{N} \right) \frac{B_o}{B_{oi}}$$

As the solution gas evolves from the oil with declining reservoir pressure, the gas saturation (assuming constant water saturation, S_{wi}) is simply given as

$$S_g = 1 - S_{wi} - S_o$$

or

$$S_g = 1 - S_{wi} - \left[(1 - S_{wi}) \left(1 - \frac{N_p}{N} \right) \frac{B_o}{B_{oi}} \right]$$

Simplifying gives

$$S_g = (1 - S_{wi}) \left[1 - \left(1 - \frac{N_p}{N} \right) \frac{B_o}{B_{oi}} \right]$$

Another important function of the MBE is history-matching the production-pressure data of individual wells. Once the reservoir pressure declines below the bubble-point pressure, it is essential to perform the following tasks:

- Generating the pseudo-relative permeability ratio, k_{rg}/k_{ro} for the entire reservoir or for individual wells' drainage areas
- Assessing the solution gas driving efficiency
- Examining the field gas-oil ratio (GOR) as compared to the laboratory solution gas solubility, R_s to define the bubble-point pressure and critical gas saturation

The instantaneous GOR, as discussed earlier, is given by:

$$GOR = \frac{Q_g}{Q_o} = R_s + \left(\frac{k_{rg}}{k_{ro}} \right) \left(\frac{\mu_o B_o}{\mu_g B_g} \right)$$

This can be arranged to solve for the relative permeability ratio, k_{rg}/k_{ro} to give:

$$\left(\frac{k_{rg}}{k_{ro}} \right) = (GOR - R_s) \left(\frac{\mu_g B_g}{\mu_o B_o} \right)$$

One of the most practical applications of the MBE is its ability to generate the field relative permeability ratio as a function of gas saturation, which can be used to adjust the laboratory core relative permeability data. *The main advantage of the field- or well-generated relative permeability ratio is that it incorporates some of the complexities of reservoir heterogeneity and degree of the segregation of the oil and the evolved gas.*

It should be noted that the laboratory relative permeability data apply to an **un-segregated** reservoir, one that has no change in fluid saturation

with height. The laboratory relative permeability is most suitable for applications with the zero-dimensional tank model. For a reservoir with complete gravity segregation, it is possible to generate a pseudo-relative permeability ratio, k_{rg}/k_{ro} . A complete segregation means that the upper part of the reservoir contains gas and immobile oil, that is, residual oil, S_{or} , while the lower part contains oil and immobile gas that exists at critical saturation, S_{gc} . Vertical communication implies that as the gas evolves in the lower region, any gas with saturation above S_{gc} moves upward rapidly and leaves that region, while in the upper region any oil above S_{or} drains downward and moves into the lower region. On the basis of these assumptions, Poston (1987) proposed the following two relationships:

$$\frac{k_{rg}}{k_{ro}} = \frac{(S_g - S_{gc})(k_{rg})_{or}}{(S_o - S_{or})(k_{ro})_{gc}}$$

$$k_{ro} = \left[\frac{S_o - S_{or}(k_{rg})_{or}}{1 - S_w - S_{gc} - S_{or}} \right] (k_{ro})_{gc}$$

where

$(k_{ro})_{gc}$ = relative permeability to oil at critical gas saturation

$(k_{go})_{or}$ = relative permeability to gas at residual oil saturation

If the reservoir is initially undersaturated (i.e., $p_i > p_b$), the reservoir pressure will continue to decline with production, and it eventually reaches the bubble-point pressure. It is recommended the material calculations be performed in two stages; first from p_i to p_b , and second from p_b to different depletion pressures p . As the pressure declines from p_i to p_b , the following changes will occur as a result:

1. Based on the water compressibility, c_w , the connate-water will expand, resulting in an increase in the connate-water saturation (provided that there is no water production)
2. Based on the formation compressibility, c_f , a reduction (compaction) in the entire reservoir pore volume

Therefore, there are several volumetric calculations that must be performed to reflect the reservoir condition at the bubble-point pressure. These calculations are based on defining the following parameters:

- Initial oil-in-place at p_i , known as N_i , with *initial* oil and water saturations of S_{oi}^i and S_{wi}^i

- Cumulative oil produced at the bubble-point pressure, N_{pb}
- Oil remaining at the bubble-point pressure, that is, *initial* oil at the bubble-point:

$$N_b = N_i - N_{pb}$$

- Total pore volume at the bubble-point pressure, $(P.V)_b$:

$(P.V)_b$ = remaining oil volume + connate-water volume + connate-water expansion – reduction in P.V due to compaction

$$(P.V)_b = (N_i - N_{pb})B_{ob} + \left[\frac{N_i B_{oi}}{1 - S_{wi}^*} \right] S_{wi}^* + \left[\frac{N_i B_{oi}}{1 - S_{wi}^*} \right] (p_i - p_b)(-c_f + c_w S_{wi}^*)$$

Simplifying gives:

$$(P.V)_b = (N_i - N_{pb})B_{ob} + \left[\frac{N_i B_{oi}}{1 - S_{wi}^*} \right] [S_{wi}^* + (p_i - p_b)(-c_f + c_w S_{wi}^*)]$$

- *Initial* oil and water saturations at the bubble-point pressure, S_{oi} and S_{wi} :

$$S_{oi} = \frac{(N_i - N_{pb})B_{ob}}{(P.V)_b} = \frac{(N_i - N_{pb})B_{ob}}{(N_i - N_{pb})B_{ob} + \left[\frac{N_i B_{oi}}{1 - S_{wi}^*} \right] [S_{wi}^* + (p_i - p_b)(-c_f + c_w S_{wi}^*)]}$$

$$S_{wi} = \frac{\left[\frac{N_i B_{oi}}{1 - S_{wi}^*} \right] [S_{wi}^* + (p_i - p_b)(-c_f + c_w S_{wi}^*)]}{(N_i - N_{pb})B_{ob} + \left[\frac{N_i B_{oi}}{1 - S_{wi}^*} \right] [S_{wi}^* + (p_i - p_b)(-c_f + c_w S_{wi}^*)]} = 1 - S_{oi}$$

- Oil saturation, S_o , at any pressure below p_b is given by:

$$S_o = \frac{(N_i - N_p)B_o}{(P.V)_b} = \frac{(N_i - N_p)B_o}{(N_i - N_{pb})B_{ob} + \left[\frac{N_i B_{oi}}{1 - S_{wi}^*} \right] [S_{wi}^* + (p_i - p_b)(-c_f + c_w S_{wi}^*)]}$$

Gas saturation, S_g at any pressure below p_b , assuming no water production, is given by:

$$S_g = 1 - S_o - S_{wi}$$

where N_i = initial oil-in-place at p_i , i.e., $p_i > p_b$, STB

N_b = initial oil-in-place at the bubble-point pressure, STB

N_{pb} = cumulative oil production at the bubble-point pressure, STB

S_{oi} = oil saturation at p_i , $p_i > p_b$

S_{oi} = initial oil saturation at p_b

S_{wi} = water saturation at p_i , $p_i > p_b$

S_{wi} = initial water saturation at p_b

It is very convenient also to qualitatively represent the fluid production graphically by employing the concept of the **bubble map**. The bubble map essentially illustrates the growing size of the drainage area of a production well. The drainage area of each well is represented by a circle with an oil bubble radius, r_{ob} , as follows:

$$r_{ob} = \sqrt{\frac{5.615 N_p}{\pi \phi h \left(\frac{1 - S_{wi}}{B_{oi}} - \frac{S_o}{B_o} \right)}}$$

This expression is based on the assumption that the saturation is evenly distributed throughout a homogeneous drainage area, where

r_{ob} = oil bubble radius, ft

N_p = well current cumulative oil production, bbl

S_o = current oil saturation

Similarly, the growing bubble of the reservoir free gas can be described graphically after calculation of the gas bubble radius, r_{gb} , of:

$$r_{gb} = \sqrt{\frac{5.615 [N R_{si} - (N - N_p) R_s - N_p R_p] B_g}{\pi \phi h (1 - S_o - S_{wi})}}$$

where r_{gb} = gas bubble radius, ft

N_p = well current cumulative oil production, bbl

B_g = current gas formation volume factor, bbl/scf

S_o = current oil saturation

Case 3. Gas-Cap-Drive Reservoirs

For a reservoir in which the expansion of the gas-cap gas is the predominant driving mechanism and assuming that the natural water influx is negligible ($W_e = 0$), the effect of water and pore compressibilities can

be considered negligible. Under these conditions, the Havlena-Odeh material balance can be expressed as:

$$F = N [E_o + m E_g] \quad (11-39)$$

where E_g is defined by Equation 11-30 as:

$$E_g = B_{oi} [(B_g/B_{gi}) - 1]$$

The way in which Equation 11-39 can be used depends on the number of unknowns in the equation. There are three possible unknowns in Equation 11-39:

- N is unknown, m is known
- m is unknown, N is known
- N and m are unknown

The practical use of Equation 11-39 in determining the three possible unknowns is presented below:

a. Unknown N , known m :

Equation 11-39 indicates that a plot of F versus $(E_o + m E_g)$ on a Cartesian scale would produce a straight line through the origin with a slope of N , as shown in Figure 11-19. In making the plot, the underground withdrawal F can be calculated at various times as a function of the production terms N_p and R_p .

Conclusion: $N = \text{Slope}$

b. Unknown m , known N :

Equation 11-39 can be rearranged as an equation of straight line, to give:

$$\left(\frac{F}{N} - E_o \right) = m E_g \quad (11-40)$$

The above relationship shows that a plot of the term $(F/N - E_o)$ versus E_g would produce a straight line with a slope of m . One advantage of this particular arrangement is that the straight line must pass through the origin which, therefore, acts as a control point. Figure 11-20 shows an illustration of such a plot.

Conclusion: $m = \text{Slope}$

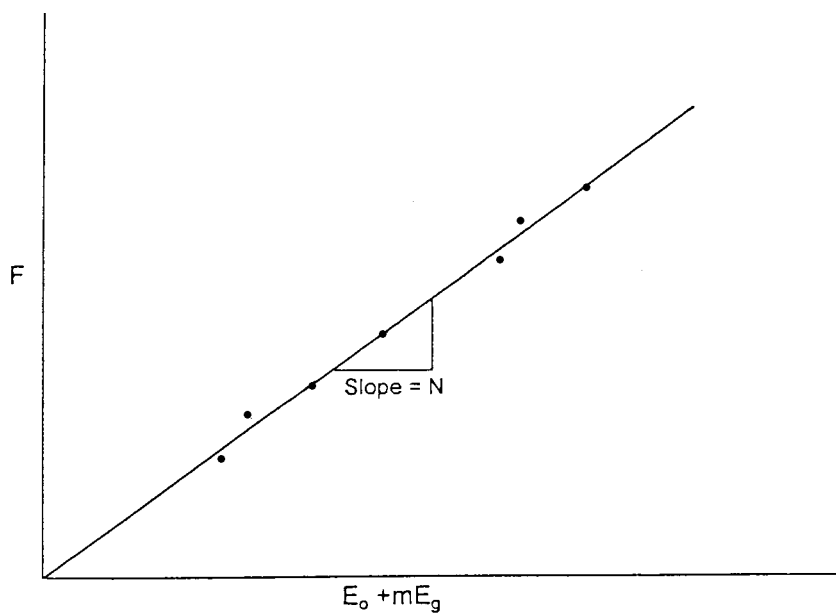


Figure 11-19. F vs. $E_o + mE_g$.

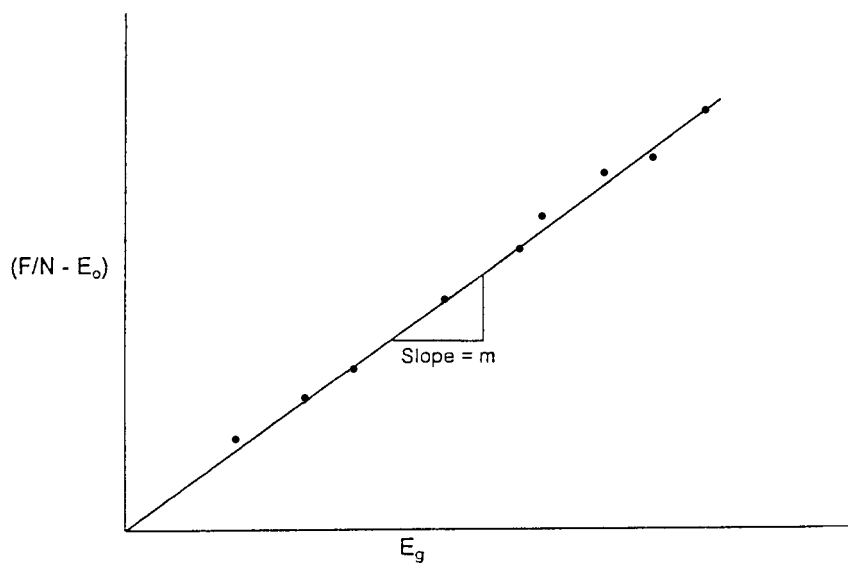


Figure 11-20. $(F/N - E_o)$ vs. E_g .

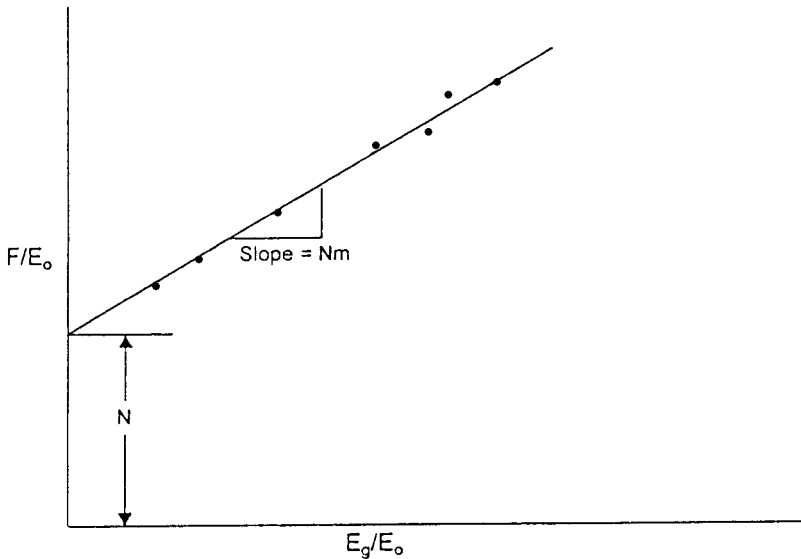


Figure 11-21. F/E_o vs. E_g/E_o .

c. N and m are Unknown

If there is uncertainty in both the values of N and m , Equation 11-39 can be re-expressed as:

$$\frac{F}{E_o} = N + mN \left(\frac{E_g}{E_o} \right) \quad (11-41)$$

A plot of F/E_o versus E_g/E_o should then be linear with intercept N and slope mN . This plot is illustrated in Figure 11-21.

Conclusions: N = Intercept
 mN = Slope
 m = Slope/intercept

Example 11-4¹

The production history and the PVT data of a gas-cap-drive reservoir are given below:

¹After Economides, M., and Hill, D., *Petroleum Production Systems*, Prentice Hall, 1993.

Date	$\bar{p}$ psi	N_p MSTB	G_p Mscf	B_t bbl/STB	B_g bbl/scf
5/1/89	4415	—	—	1.6291	0.00077
1/1/91	3875	492.5	751.3	1.6839	0.00079
1/1/92	3315	1015.7	2409.6	1.7835	0.00087
1/1/93	2845	1322.5	3901.6	1.9110	0.00099

The initial gas solubility R_{si} is 975 scf/STB. Estimate the initial oil and gas-in-place.

Solution

Step 1. Calculate the cumulative produced gas-oil ratio R_p

$\bar{p}$	G_p Mscf	N_p MSTB	$R_p = G_p/N_p$ scf/STB
4415	—	—	—
3875	751.3	492.5	1525
3315	2409.6	1015.7	2372
2845	3901.6	1322.5	2950

Step 2. Calculate F , E_o , and E_g :

p	F	E_o	E_g
3875	2.04×10^6	0.0548	0.0529
3315	8.77×10^6	0.1540	0.2220
2845	17.05×10^6	0.2820	0.4720

Step 3. Calculate F/E_o and E_g/E_o

p	F/E_o	E_g/E_o
3875	3.72×10^7	0.96
3315	5.69×10^7	1.44
2845	6.00×10^7	1.67

Step 4. Plot (F/E_o) versus (E_g/E_o) as shown in Figure 11-22 to give:

- Intercept = $N = 9$ MMSTB
- Slope = $N m = 3.1 \times 10^7$

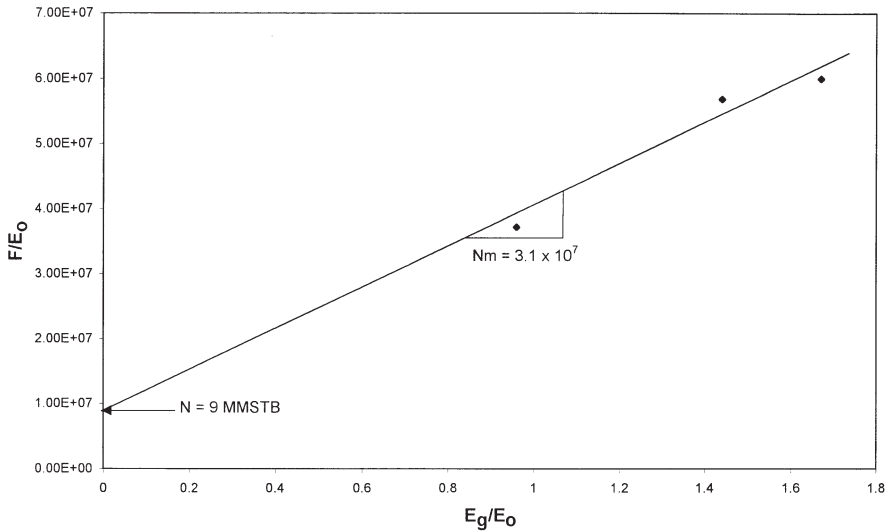


Figure 11-22. Calculation of m and N for Example 11-4.

Step 5. Calculate m :

$$m = 3.1 \times 10^7 / (9 \times 10^6) = 3.44$$

Step 6. Calculate initial gas-in-place G :

$$m = \frac{GB_{gi}}{NB_{oi}}$$

$$G = \frac{(3.44)(9 \times 10^6)(1.6291)}{0.00077} = 66 \text{ MMMscf}$$

Case 4. Water-Drive Reservoirs

In a water-drive reservoir, identifying the type of the aquifer and characterizing its properties are perhaps the most challenging tasks involved in conducting a reservoir engineering study. Yet, without an accurate description of the aquifer, future reservoir performance and management cannot be properly evaluated.

The full MBE can be expressed again as:

$$F = N(E_o + m E_g + E_{f,w}) + W_e$$

Dake (1978) points out that the term $E_{f,w}$ can frequently be neglected in water-drive reservoirs. This is not only for the usual reason that the water and pore compressibilities are small, but also because a water influx helps to maintain the reservoir pressure and, therefore, the Δp appearing in the $E_{f,w}$ term is reduced, or

$$F = N (E_o + m E_g) + W_e \quad (11-42)$$

If, in addition, the reservoir has an initial gas cap, then Equation 11-42 can be further reduced to:

$$F = N E_o + W_e \quad (11-43)$$

Dake (1978) points out that in attempting to use the above two equations to match the production and pressure history of a reservoir, the greatest uncertainty is always the determination of the water influx W_e . In fact, in order to calculate the influx the engineer is confronted with what is inherently the greatest uncertainty in the whole subject of reservoir engineering. The reason is that the calculation of W_e requires a mathematical model which itself relies on the knowledge of aquifer properties. These, however, are seldom measured since wells are not deliberately drilled into the aquifer to obtain such information.

For a water-drive reservoir with no gas cap, Equation 11-43 can be rearranged and expressed as:

$$\frac{F}{E_o} = N + \frac{W_e}{E_o} \quad (11-44)$$

Several water influx models have been described in Chapter 10, including the:

- Pot-aquifer model
- Schilthuis steady-state method
- Van Everdingen-Hurst model

The use of these models in connection with Equation 11-44 to simultaneously determine N and W_e is described below.

The Pot-Aquifer Model in the MBE

Assume that the water influx could be properly described using the simple pot-aquifer model given by Equation 10-5 as:

$$W_e = (c_w + c_f) W_i f (p_i - p) \quad (11-45)$$

$$f = \frac{(\text{encroachment angle})^\circ}{360^\circ} = \frac{\theta}{360^\circ}$$

$$W_i = \left[\frac{\pi(r_a^2 - r_e^2) h \phi}{5.615} \right]$$

where r_a = radius of the aquifer, ft
 r_e = radius of the reservoir, ft
 h = thickness of the aquifer, ft
 ϕ = porosity of the aquifer
 θ = encroachment angle
 c_w = aquifer water compressibility, psi^{-1}
 c_f = aquifer rock compressibility, psi^{-1}
 W_i = initial volume of water in the aquifer, bbl

Since the aquifer properties c_w , c_f , h , r_a , and θ are seldom available, it is convenient to combine these properties and treat as one unknown K . Equation 11-45 can be rewritten as:

$$W_e = K \Delta p \quad (11-46)$$

Combining Equation 11-46 with Equation 11-44 gives:

$$\frac{F}{E_o} = N + K \left(\frac{\Delta p}{E_o} \right) \quad (11-47)$$

Equation 11-47 indicates that a plot of the term (F/E_o) as a function of $(\Delta p/E_o)$ would yield a straight line with an intercept of N and slope of K , as illustrated in Figure 11-23.

The Steady-State Model in the MBE

The steady-state aquifer model as proposed by Schilthuis (1936) is given by:

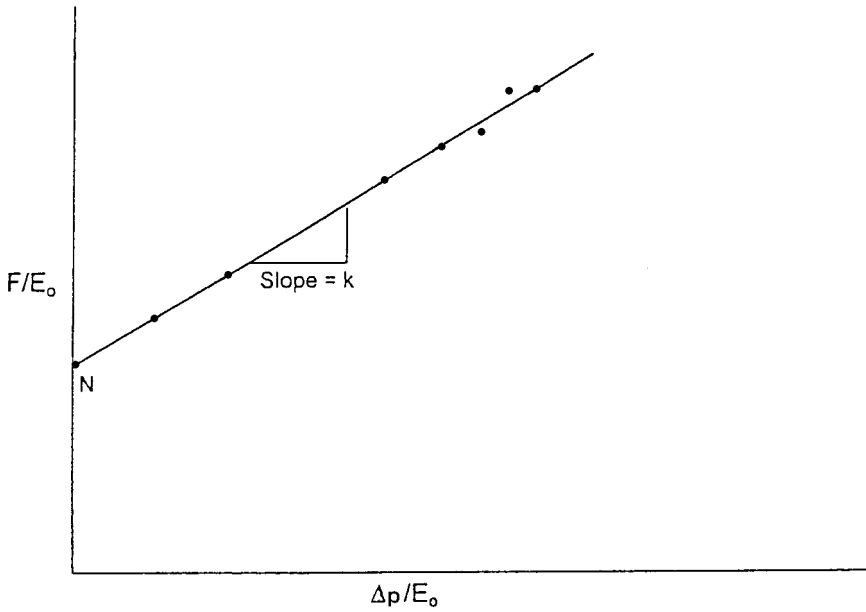


Figure 11-23. F/E_o vs. $\Delta p/E_o$.

$$W_e = C \int_0^t (p_i - p) dt \quad (11-48)$$

where W_e = cumulative water influx, bbl

C = water influx constant, bbl/day/psi

t = time, days

p_i = initial reservoir pressure, psi

p = pressure at the oil-water contact at time t , psi

Combining Equation 11-48 with Equation 11-44 gives:

$$\frac{F}{E_o} = N + C \left(\frac{\int_0^t (p_i - p) dt}{E_o} \right) \quad (11-49)$$

Plotting (F/E_o) versus $\int_0^t (p_i - p) dt/E_o$ results in a straight line with an intercept that represents the initial oil-in-place N and a slope that describes the water influx C as shown in Figure 11-24.

The Unsteady-State Model in the MBE

The van Everdingen-Hurst unsteady-state model is given by:

$$W_e = B \sum \Delta p W_{eD} \quad (11-50)$$

with

$$B = 1.119 \phi c_t r_e^2 h f$$

Van Everdingen and Hurst presented the dimensionless water influx W_{eD} as a function of the dimensionless time t_D and dimensionless radius r_D that are given by:

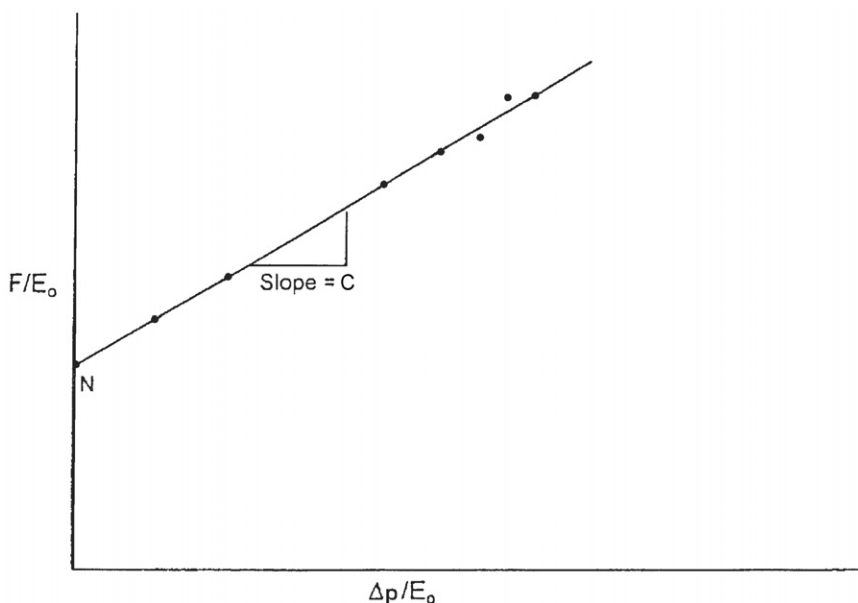


Figure 11-24. Graphical determination of N and C .

$$t_D = 6.328 \times 10^{-3} \frac{kt}{\phi \mu_w c_t r_e^2}$$

$$r_D = \frac{r_a}{r_e}$$

$$c_t = c_w + c_f$$

where t = time, days

k = permeability of the aquifer, md

ϕ = porosity of the aquifer

μ_w = viscosity of water in the aquifer, cp

r_a = radius of the aquifer, ft

r_e = radius of the reservoir, ft

c_w = compressibility of the water, psi^{-1}

Combining Equation 11-50 with Equation 11-44 gives:

$$\frac{F}{E_o} = N + B \left(\frac{\sum \Delta p W_{eD}}{E_o} \right) \quad (11-51)$$

The proper methodology of solving the above linear relationship is summarized in the following steps.

Step 1. From the field past production and pressure history, calculate the underground withdrawal F and oil expansion E_o .

Step 2. Assume an aquifer configuration, i.e., linear or radial.

Step 3. Assume the aquifer radius r_a and calculate the dimensionless radius r_D .

Step 4. Plot (F/E_o) versus $(\sum \Delta p W_{eD})/E_o$ on a Cartesian scale. If the assumed aquifer parameters are correct, the plot will be a straight line with N being the intercept and the water influx constant B being the slope. It should be noted that four other different plots might result. These are:

- Complete random scatter of the individual points, which indicates that the calculation and/or the basic data are in error.

- A systematically upward curved line, which suggests that the assumed aquifer radius (or dimensionless radius) is too small.
- A systematically downward curved line, indicating that the selected aquifer radius (or dimensionless radius) is too large.
- An s-shaped curve indicates that a better fit could be obtained if a linear water influx is assumed.

Figure 11-25 shows a schematic illustration of Havlena-Odeh (1963) methodology in determining the aquifer fitting parameters.

Example 11-5

The material balance parameters, the underground withdrawal F , and oil expansion E_o of a saturated-oil reservoir (i.e., $m = 0$) are given below:

p	F	E_o
3500	—	—
3488	2.04×10^6	0.0548
3162	8.77×10^6	0.1540
2782	17.05×10^6	0.2820

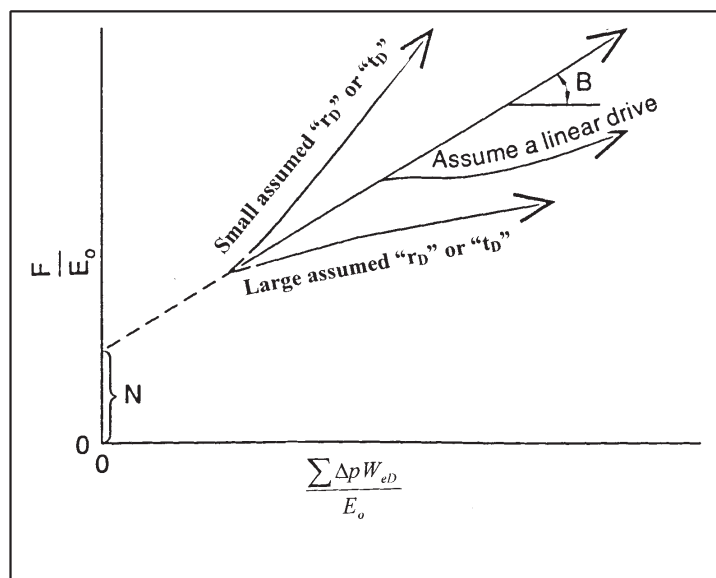


Figure 11-25. Havlena and Odeh straight-line plot. (After Havlena and Odeh, 1963.)

Assuming that the rock and water compressibilities are negligible, calculate the initial oil-in-place.

Solution

Step 1. The most important step in applying the MBE is to verify that no water influx exists. Assuming that the reservoir is volumetric, calculate the initial oil-in-place N by using every individual production data point in Equation 11-38, or:

$$N = F/E_o$$

F	E_o	$N = F/E_o$
2.04×10^6	0.0548	37 MMSTB
8.77×10^6	0.1540	57 MMSTB
17.05×10^6	0.2820	60 MMSTB

Step 2. The above calculations show the calculated values of the initial oil-in-place are increasing (as shown graphically in Figure 11-26), which indicates a water encroachment, i.e., water-drive reservoir.

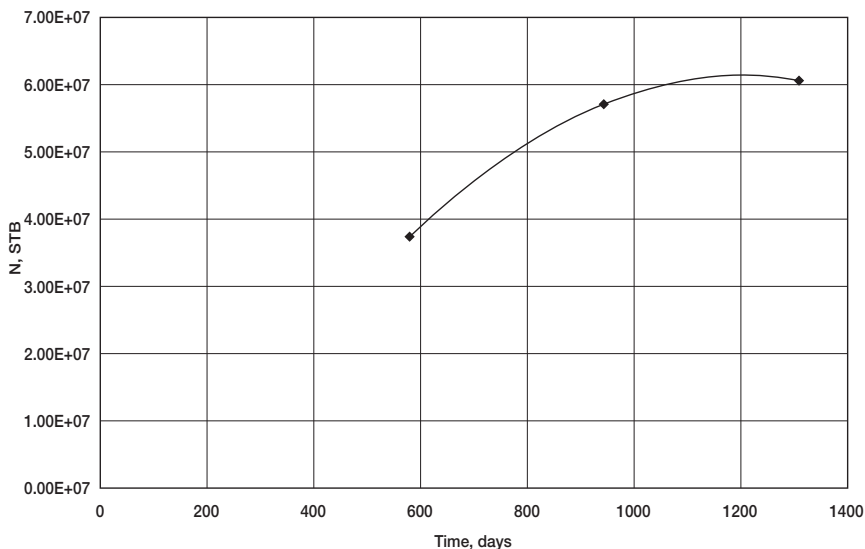


Figure 11-26. Indication of water influx.

Step 3. For simplicity, select the pot-aquifer model to represent the water encroachment calculations in the MBE as given by Equation 11-47, or:

$$\frac{F}{E_o} = N + K \left(\frac{\Delta p}{E_o} \right)$$

Step 4. Calculate the terms (F/E_o) and (Δp/E_o) of Equation 11-47.

p	Δp	F	E _o	F/E _o	Δp/E _o
3500	0	—	—	—	—
3488	12	2.04 × 10 ⁶	0.0548	37.23 × 10 ⁶	219.0
3162	338	8.77 × 10 ⁶	0.1540	56.95 × 10 ⁶	2194.8
2782	718	17.05 × 10 ⁶	0.2820	60.46 × 10 ⁶	2546

Step 5. Plot (F/E_o) versus (Δp/E_o), as shown in Figure 11-27, and determine the intercept and the slope.

Intercept = N = 35 MMSTB

Slope = K = 9983

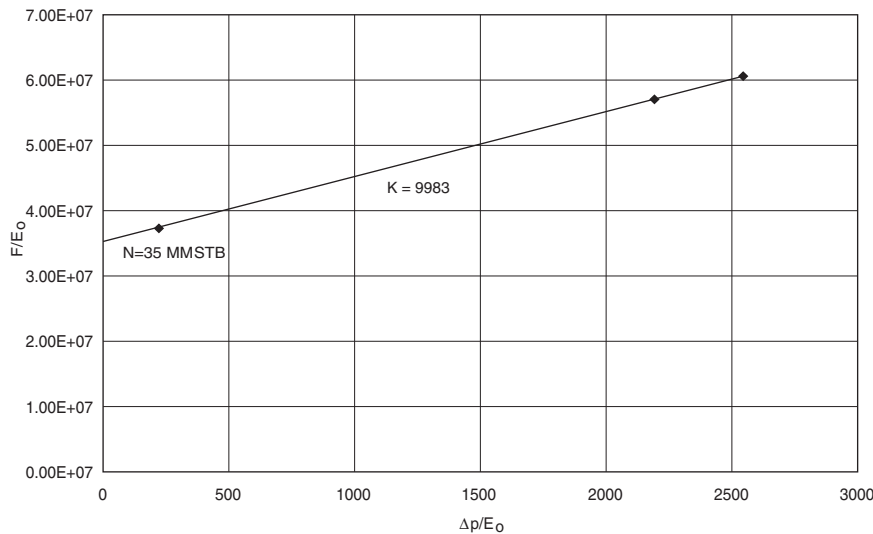


Figure 11-27. F/E_o versus Δp/E_o.

Case 5. Combination-Drive Reservoirs

This relatively complicated case involves the determination of the following three unknowns:

- Initial oil-in-place, N
- Size of the gas cap, m
- Water influx, W_e

The general MBE that includes these three unknowns is given by Equation 11-32:

$$F = N (E_o + mE_g) + W_e$$

Where the variables constituting the above expressions are defined by

$$\begin{aligned} F &= N_p [B_o + (R_p - R_s) B_g] + W_p B_w \\ &= N_p [B_t + (R_p - R_s) B_g] + W_p B_w \end{aligned}$$

$$\begin{aligned} E_o &= (B_o + B_{oi}) + (R_{si} - R_s) B_g \\ &= B_t + B_{ti} \end{aligned}$$

$$E_g = B_{oi} [(B_g/B_{gi}) - 1]$$

Havlena and Odeh differentiated Equation 11-32 with respect to pressure and rearranged the resulting equation to eliminate m , to give:

$$\frac{F E_g' - F' E_g}{E_o E_g' - E_o' E_g} = N + \frac{W_e E_g' - W_e' E_g}{E_o E_g' - E_o' E_g}$$

in which the primes denote derivatives with respect to pressure, that is,

$$E_g' = \frac{\partial E_g}{\partial p} = \left(\frac{B_{oi}}{B_{gi}} \right) \frac{\partial B_g}{\partial p} \approx \left(\frac{B_{oi}}{B_{gi}} \right) \frac{\Delta B_g}{\Delta p}$$

$$E_o' = \frac{\partial E_o}{\partial p} = \frac{\partial B_t}{\partial p} \approx \frac{\Delta B_t}{\Delta p}$$

$$F' = \frac{\partial F}{\partial p} \approx \frac{\Delta F}{\Delta p}$$

$$W_e' = \frac{\partial W_e}{\partial p}$$

A plot of the left-hand side of the equation versus the second term on the right for a selected aquifer model should, if the choice is correct, provide a straight line with unit slope whose intercept on the ordinate gives the initial oil-in-place, N . Having correctly determined N and W_e , Equation 11-32 can be solved directly for m , to give:

$$m = \frac{F - N E_o - W_e}{N E_g}$$

Notice that all the given derivatives can be evaluated numerically using one of the finite difference techniques: forward, backward, or central difference formula.

Case 6. Average Reservoir Pressure

To gain an understanding of the behavior of a reservoir with free gas, for example, solution gas drive or gas-cap drive, it is essential that every effort be made to determine reservoir pressures with accuracy. In the absence of reliable pressure data, the MBE can be used to estimate average reservoir pressure if accurate values of m and N are available from volumetric calculations. The general MBE is given by Equation 11-39 as:

$$F = N[E_o + m E_g]$$

Solving Equation 11-39 for the average pressure using the production history of the field involves the following graphical procedure:

- Step 1.* Select the time at which the average reservoir pressure is to be determined and obtain the corresponding production data, N_p , G_p , and R_p .
- Step 2.* Assume several average reservoir pressure values and determine the left-hand side, F , of Equation 11-39 at each assumed pressure:

$$F = N_p [B_o + (R_p - R_s) B_g] + W_p B_w$$

- Step 3.* Using the same assumed average reservoir pressure values as in Step 2, calculate the right-hand side of Equation 11-39:

$$\text{RHS} = N[E_o + m E_g]$$

where

$$E_o = (B_o - B_{oi}) + (R_{si} - R_s) B_g$$

$$E_g = B_{oi} [(B_g/B_{gi}) - 1]$$

Step 4. Plot the left-hand and right-hand sides of the MBE as calculated in Steps 2 and 3, on Cartesian paper, as a function of assumed average pressure. The point of intersection gives the average reservoir pressure that corresponds to the selected time of Step 1.

Step 5. Repeat Steps 1 through 4 to estimate reservoir pressure at each selected depletion time.

Tracy's Form of the Material Balance Equation

Neglecting the formation and water compressibilities, the general material balance equation as expressed by Equation 11-13 can be reduced to the following:

$$N = \frac{N_p B_o + (G_p - N_p R_s) B_g - (W_e - W_p B_w)}{(B_o - B_{oi}) + (R_{si} - R_s) B_g + m B_{oi} \left[\frac{B_g}{B_{gi}} - 1 \right]} \quad (11-52)$$

Tracy (1955) suggested that the above relationship can be rearranged into a more usable form as:

$$N = N_p \Phi_o + G_p \Phi_g + (W_p B_w - W_e) \Phi_w \quad (11-53)$$

where Φ_o , Φ_g , and Φ_w are considered PVT related properties that are functions of pressure and defined by:

$$\Phi_o = \frac{B_o - R_s B_g}{\text{Den}} \quad (11-54)$$

$$\Phi_g = \frac{B_g}{\text{Den}} \quad (11-55)$$

$$\phi_w = \frac{1}{\text{Den}} \quad (11-56)$$

with

$$\text{Den} = (B_o - B_{oi}) + (R_{si} - R_s)B_g + m B_{oi} \left[\frac{B_g}{B_{gi}} - 1 \right] \quad (11-57)$$

where Φ_o = oil PVT function

Φ_g = gas PVT function

Φ_w = water PVT function

Figure 11-28 gives a graphical presentation of the behavior of Tracy's PVT functions with changing pressure.

Notice that Φ_o is negative at low pressures and all Φ functions are approaching infinity at bubble-point pressure. Tracy's form is valid only for initial pressures equal to bubble-point pressure and cannot be used at pressures above bubble-point. Furthermore, the shape of the Φ function curves illustrates that small errors in pressure and/or production can cause large errors in calculated oil-in-place at pressures near the bubble-point.

Steffensen (1992), however, pointed out that Tracy's equation uses the oil formation volume factor at the bubble-point pressure B_{ob} for the initial B_{oi} , which causes all the PVT functions to become infinity at the bubble-point pressure. Steffensen suggested that Tracy's equation could be extended for applications above the bubble-point pressure, i.e., for undersaturated-oil reservoirs, by simply using the value of B_o at the initial reservoir pressure. He concluded that Tracy's methodology could predict reservoir performance for the entire pressure range from any initial pressure down to abandonment.

The following example is given by Tracy (1955) to illustrate his proposed approach.

Example 11-6

The production history of a saturated-oil reservoir is as follows:

Pressure, psia	Cumulative Oil, MSTB	Cumulative Gas, MMscf
1690	0	0
1600	398	38.6
1500	1570	155.8
1100	4470	803

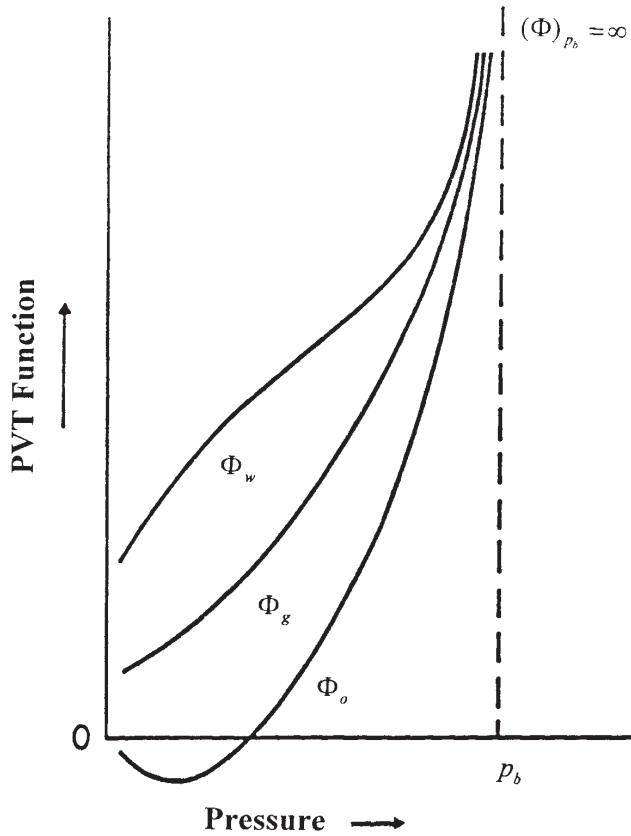


Figure 11-28. Tracy's PVT functions.

The calculated values of the PVT functions are given below:

Pressure, psia	Φ_o	Φ_g
1600	36.60	0.4000
1500	14.30	0.1790
1100	2.10	0.0508

Calculate the oil-in-place N .

Solution

The calculations can be conveniently performed in following table:

p , psia	N_p , MSTB	G_p , MMscf	$(N_p \Phi_o)$	$(G_p \Phi_g)$	N , STB
1600	398	38.6	14.52×10^6	15.42×10^6	29.74×10^6
1500	155.8	155.8	22.45×10^6	27.85×10^6	50.30×10^6
1100	803.0	803.0	9.39×10^6	40.79×10^6	50.18×10^6

The above results show that the original oil-in-place in this reservoir is approximately 50 MMSTB of oil. The calculation at 1600 psia is a good example of the sensitivity of such a calculation near the bubble-point pressure. Since the last two values of the original oil-in-place agree so well, the first calculation is probably wrong.

PROBLEMS

- Given the following data on an oil reservoir:

	Oil	Aquifer
Geometry	circle	semi-circle
Encroachment angle	—	180°
Radius, ft	4000	80,000
Flow regime	semisteady-state	unsteady-state
Porosity	—	0.20
Thickness, ft	—	30
Permeability, md	200	50
Viscosity, cp	1.2	0.36
Original pressure	3800	3800
Current pressure	3600	—
Original volume factor	1.300	1.04
Current volume factor	1.303	1.04
Bubble-point pressure	3000	—

The field has been on production for 1120 days and has produced 800,000 STB of oil and 60,000 STB of water. Water and formation compressibilities are estimated to 3×10^{-6} and 3.5×10^{-6} psi^{-1} , respectively. Calculate the original oil-in-place.

- The following rock- and fluid-properties data are available on the Nameless Fields:

Reservoir area = 1000 acres porosity = 10% thickness = 20'

$$T = 140^{\circ}\text{F}$$

$$s_{wi} = 20\%$$

$$p_i = 4000 \text{ psi}$$

$$p_b = 4000 \text{ psi}$$

The gas compressibility factor and relative permeability ratio are given by the following expressions:

$$z = 0.8 - 0.00002 (p - 4000)$$

$$\frac{k_{rg}}{k_{ro}} = 0.00127 e^{17.269 S_g}$$

The production history of the field is given below:

	4000 psi	3500 psi	3000 psi
μ_o , cp	1.3	1.25	1.2
μ_g , cp	—	0.0125	0.0120
B_o , bbl/STB	1.4	1.35	1.30
R_s , scf/STB	—	—	450
GOR, scf/STB	600	—	1573

Subsurface information indicates that there is no aquifer and has been no water production.

Calculate:

- Remaining oil-in-place at 3,000 psi
 - Cumulative gas produced at 3,000 psi
3. The following PVT and production history data are available on an oil reservoir in West Texas:

Original oil-in-place = 10 MMSTB

Initial water saturation = 22%

Initial reservoir pressure = 2496 psia

Bubble-point pressure = 2496 psi

Pressure psi	B_o bbl/STB	R_s scf/STB	B_g bbl/scf	μ_o cp	μ_g cp	GOR scf/STB
2496	1.325	650	0.000796	0.906	0.016	650
1498	1.250	486	0.001335	1.373	0.015	1360
1302	1.233	450	0.001616	1.437	0.014	2080

The cumulative gas-oil ratio at 1,302 psi is recorded at 953 scf/STB. Calculate:

- a. Oil saturation at 1,302 psia
 - b. Volume of the free gas in the reservoir at 1,302 psia
 - c. Relative permeability ratio (k_g/k_o) at 1,302 psia
4. The Nameless Field is an undersaturated-oil reservoir. The crude oil system and rock type indicate that the reservoir is highly compressible. The available reservoir and production data are given below:

$$S_{wi} = 0.25 \quad \phi = 20\% \quad \text{Area} = 1000 \text{ acres}$$

$$h = 70' \quad T = 150^\circ\text{F}$$

Bubble-point pressure = 3500 psia

	Original Condition	Current Conditions
Pressure, psi	5000	4500
B_o , bbl/STB	1.905	1.920
R_s , scf/STB	700	700
N_p , MSTB	0	610.9

Calculate the cumulative oil production at 3,900 psi. The PVT data show that the oil formation volume factor is equal to 1.938 bbl/STB at 3,900 psia.

5. The following data² are available on a gas-cap-drive reservoir:

Pressure (psi)	N_p (MMSTB)	R_p (scf/STB)	B_o (RB/STB)	R_s (scf/STB)	B_g (RB/scf)
3330			1.2511	510	0.00087
3150	3.295	1050	1.2353	477	0.00092
3000	5.903	1060	1.2222	450	0.00096
2850	8.852	1160	1.2122	425	0.00101
2700	11.503	1235	1.2022	401	0.00107
2550	14.513	1265	1.1922	375	0.00113
2400	17.730	1300	1.1822	352	0.00120

Calculate the initial oil and free gas volumes.

6. The Wildcat Reservoir was discovered in 1980. This reservoir had an initial reservoir pressure of 3,000 psia, and laboratory data indicated a bubble-point pressure of 2,500 psi. The following additional data are available:

²Dake, L. P., *Fundamentals of Reservoir Engineering*, Elsevier Publishing Co., Amsterdam, 1978.

Area	= 700 acres
Thickness	= 35 ft
Porosity	= 20%
Temperature	= 150°F
API gravity	= 50°
Specific gravity of gas	= 0.72
Initial water saturation	= 25%

Average isothermal oil compressibility above the bubble point = $18 \times 10^{-6} \text{ psi}^{-1}$

Calculate the volume of oil initially in place at 3,000 psi as expressed in STB.

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